

Queimadas Reference Satellite: A data analysis

Analysing data for choosing the next reference satellite for the Queimadas Program

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October 1, 2026



Agenda I

- 1 Introduction
- 2 Exploratory Data Analysis
- 3 Time Series
 - Time series decomposition
 - Reference satellites
 - Breakpoint identification
- 4 *Queimadas* method (12-month regression)
 - Regression
 - Forecast
- 5 *Queimadas* method (1-month regression)
 - Regression
 - Forecast
 - Forecast differences (12 versus 1 month)
- 6 Future work
- 7 References



Introduction



What's new?

- Queimadas method:

- ▶ Updated *Queimadas* database to include data from 1998. 
- ▶ BFast:
 - ★ AQUA_M-T still breaks on 2007-11-15. 
 - ★ New break on NOAA-12 on 2005-11-15. 



What is a reference satellite?

The reference satellite is [3]:

- It's the satellite that daily senses fire foci used to build time series.
- NOAA-12 (AVHRR sensor, late afternoon pass) from 1998-06-01 to 2002-07-03.
- Then, AQUA_M-T (MODIS sensor, early afternoon pass).
- The reference satellite senses only a fraction of the total fire foci.
- Evenso, this allows trend analysis.
- Next reference satellite, NPP-SUOMI (VIIRS sensor).



What would be the next reference satellite?

- What would be the next reference satellite [2, 4]?



Online resources

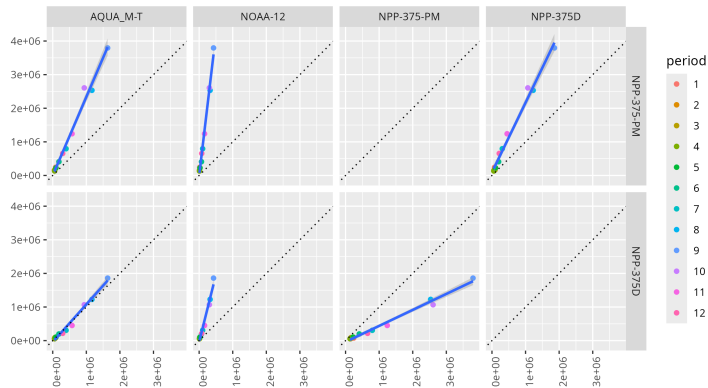
- Queimadas [portal](#).
- Queimadas [FAQ](#).
- Source code
 - ▶ Code repository available [here](#).
 - ▶ Slides available [here](#).



Exploratory Data Analysis



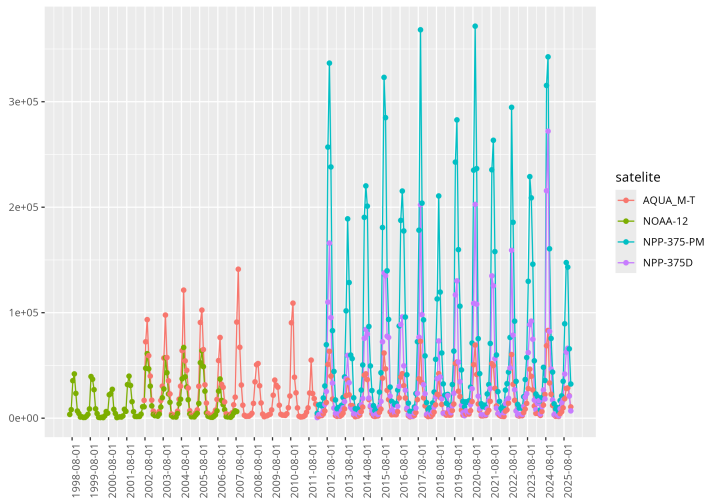
Brazilian fire foci aggregated by month (all years)



VIIRS detects more fire foci than AQUA M-T and NOAA 12.



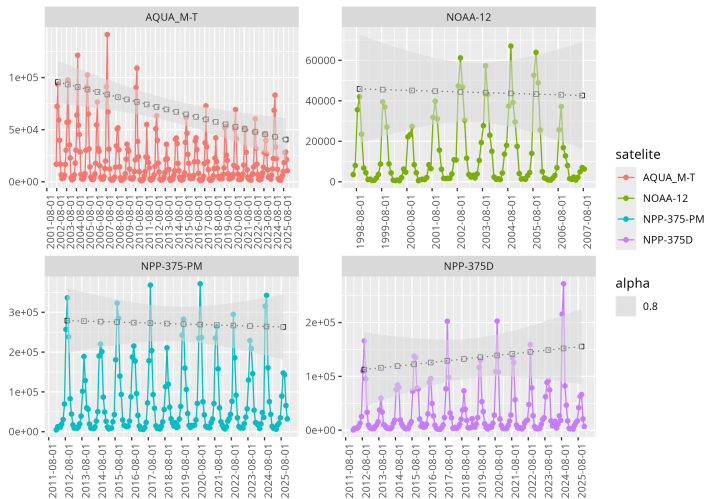
Brazilian fire foci aggregated by year and month



AQUA M-T has the longest time series of fire foci.



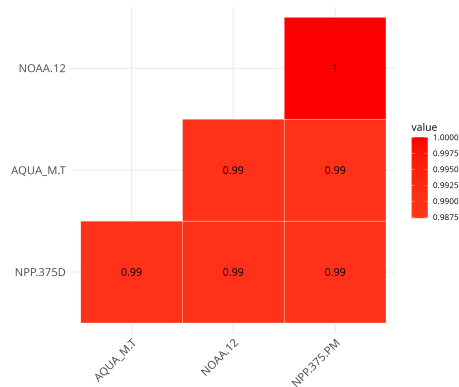
Brazilian fire foci aggregated by year and month (facets)



AQUA M-T and NOAA-12 have a annual-peak negative trend.

Correlation between fire foci from the reference satellites in Brazil

- The time series overlap partially.
- Use only the month (not the year) to compute correlation.
- This way we can use the whole time series in the estimation.



There are high correlation among fire foci time series.



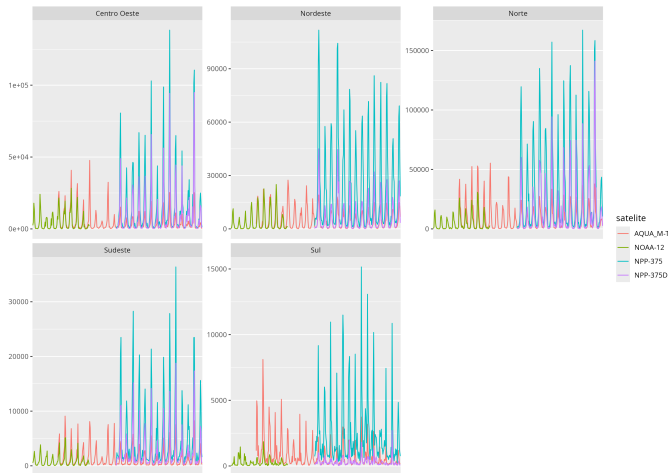
Correlation between fire foci from the reference satellites in Brazil

- The time series overlap partially.
- Estimate the correlation using only observation that overlap.
- Some satellites don't overlap and their correlation can't be estimated.

satellite_x	satellite_y	cor_val
NOAA-12	AQUA_M-T	0.928
AQUA_M-T	NPP-375-PM	0.978
AQUA_M-T	NPP-375D	0.962
NPP-375-PM	NPP-375D	0.958

Table: High correlation among fire foci time series.

Brazilian regions by year and month



Fire foci time series by brazilian region.

Time Series

Time series' components

- Structural time series model [1]:

- ▶ Trend $g(t)$.
- ▶ Periodic changes $s(t)$.
- ▶ Effects of holidays $h(t)$.

- **Assumptions:**

- ▶ ϵ_t is normally distributed.
- ▶ Forecasting is a curve-fitting problem; no account for temporal dependence structure in the data [5].

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

Do the models fit?

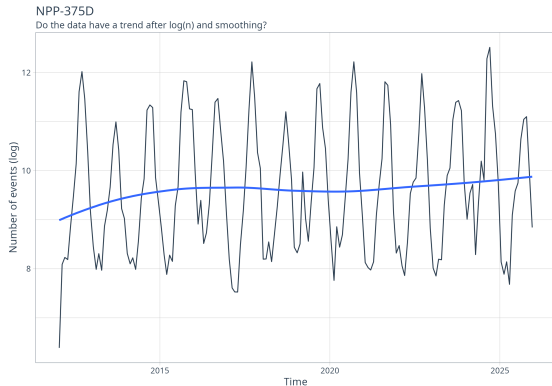
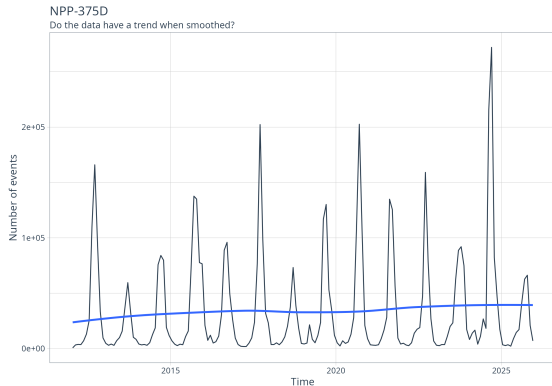
- Do the data have a trend when smoothed?
- Do the data have a trend after applying the logarithm function and smoothing?
- Are the data autocorrelated in the time dimension?
- Do the data have a trend component?
- Do the data have season component?
- How does the model fit its observations?
- How do compare the residuals and the fitted values?
- Do the mode-observations residual have a normal distribution?



NPP-375D

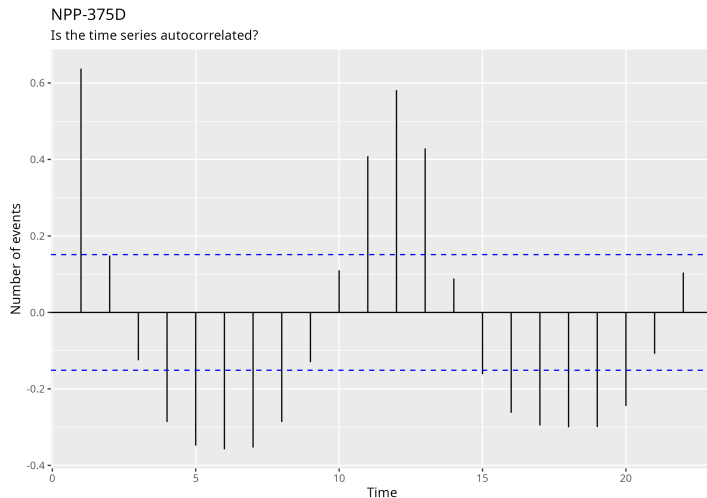


Time series trend (with smoothing)



Do the data have a trend when smoothed? (data and data log to the left).

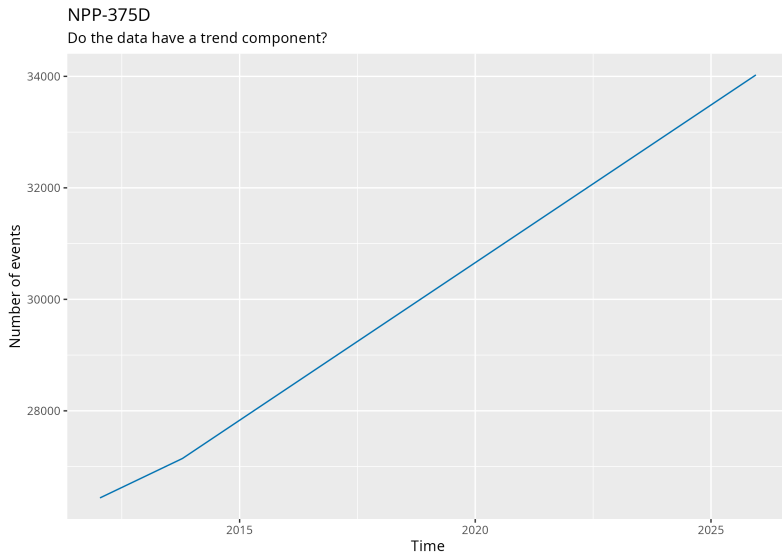
Time series autocorrelation



Is the time series autocorrelated?



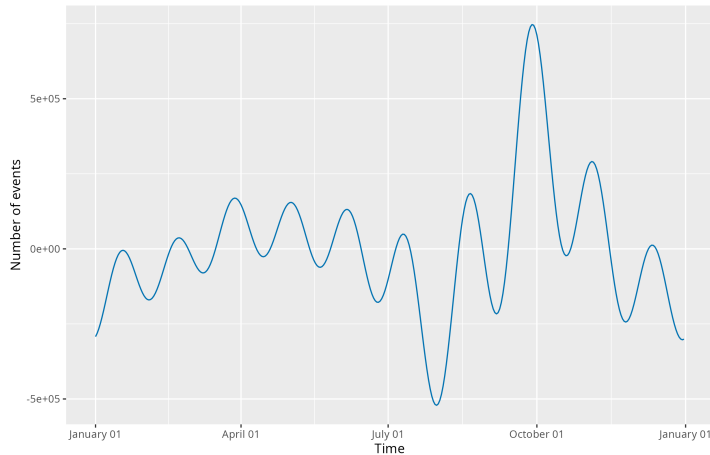
Time series trend



Time series seasonality

NPP-375D

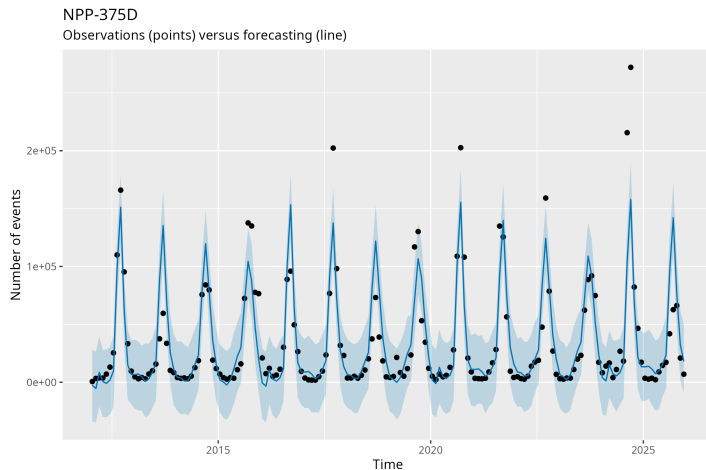
Do the data have a seasonality component?



Do the data have a seasonality component?

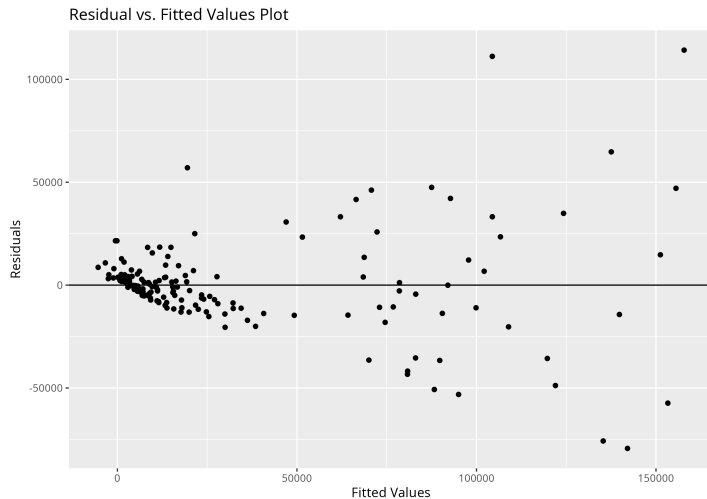


Time series versus forecast



Observations (black dots) and model (blue line) with a 95% confidence interval (gray shadow)

Time series versus forecast



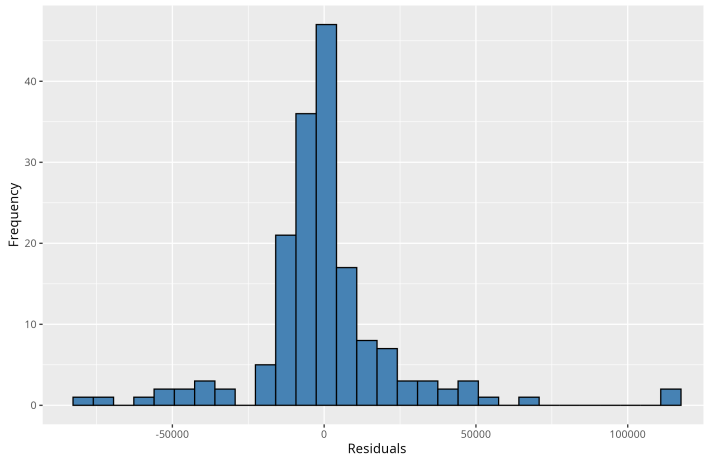
Large observations of fire foci have larger residuals.



Time series versus forecast

Histogram of Residuals

Do the residuals have a normal distribution?



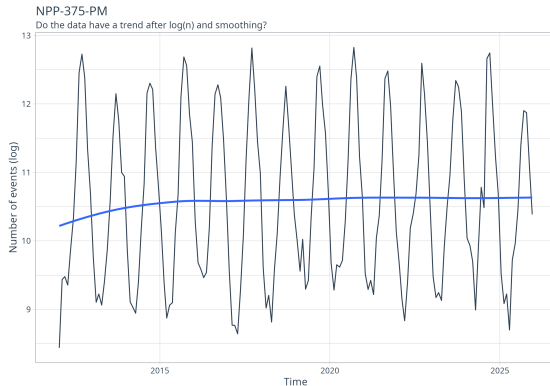
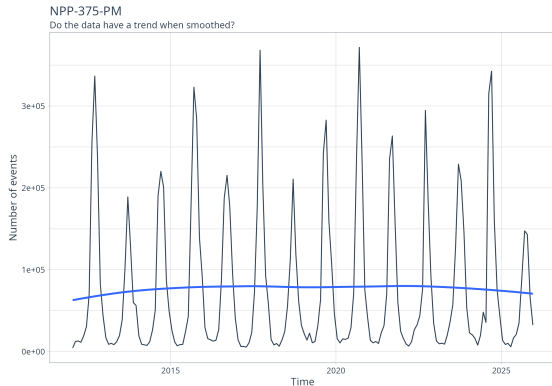
Do the residuals have a normal distribution?



NPP-375-PM

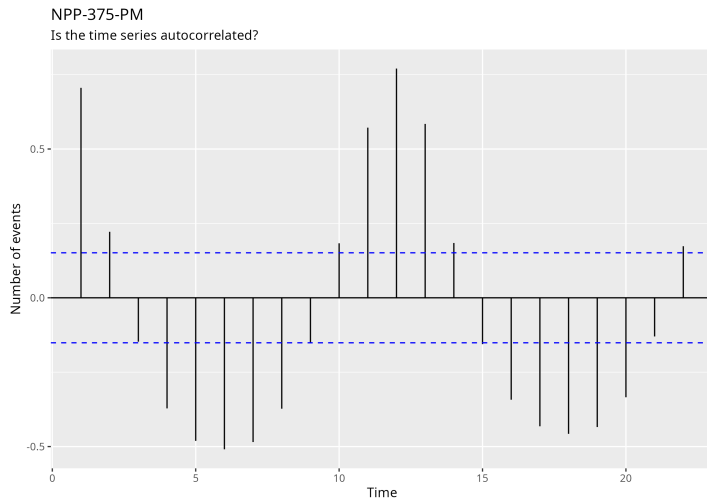


Time series trend (with smoothing)



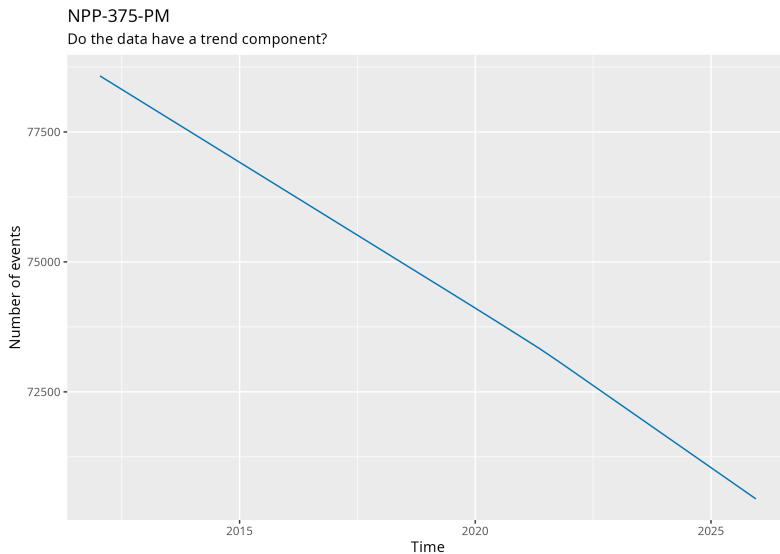
Do the data have a trend when smoothed?

Time series autocorrelation



Is the time series autocorrelated?

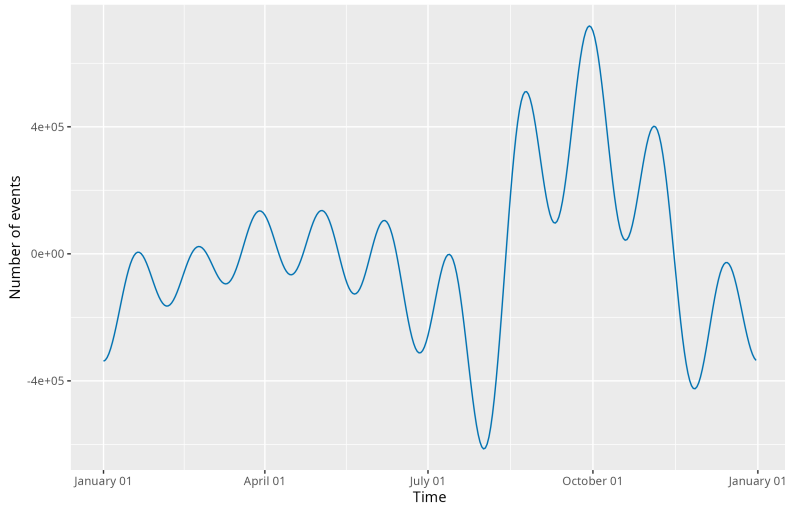
Time series trend



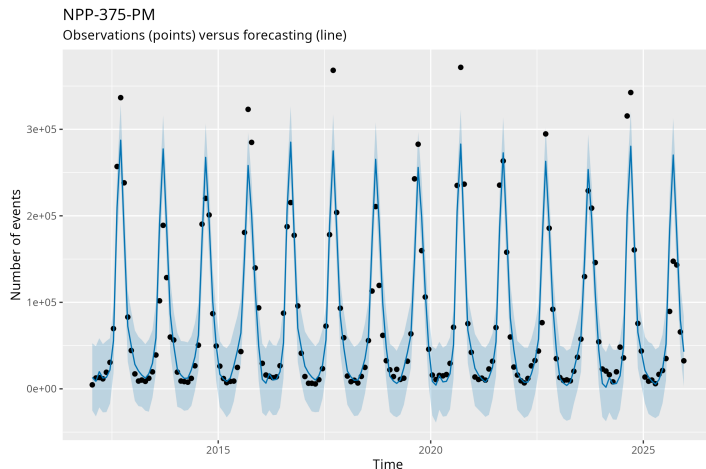
Time series seasonality

NPP-375-PM

Do the data have a seasonality component?

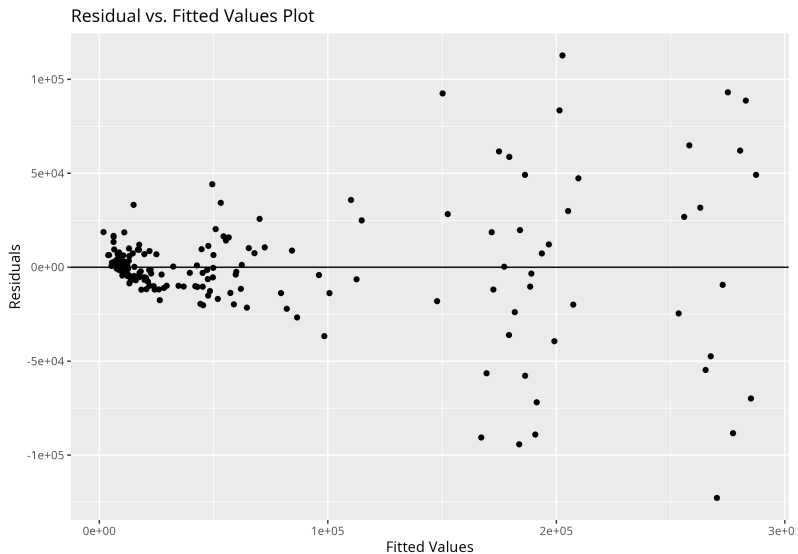


Time series versus forecast



Observations (black dots) and model (blue line) with a confidence interval of 95% (gray shadow)

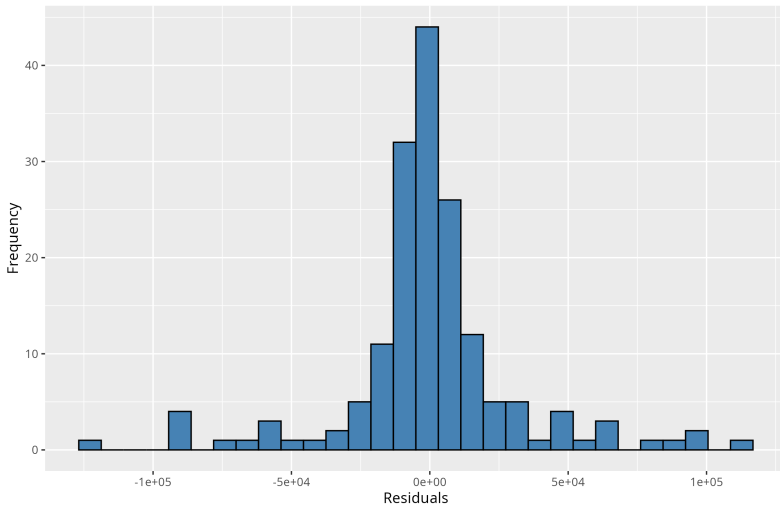
Time series versus forecast



Time series versus forecast

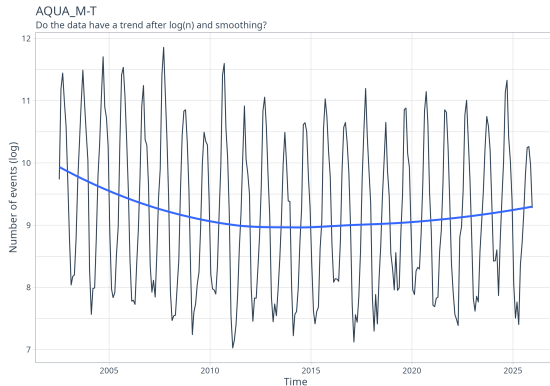
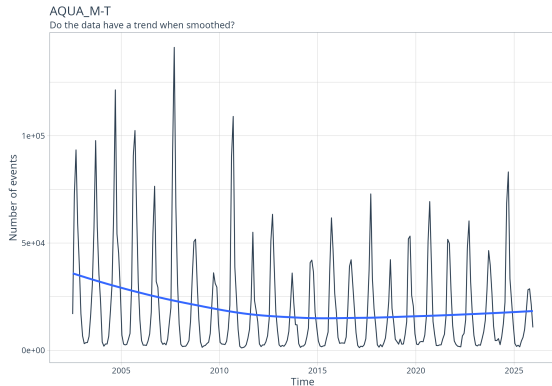
Histogram of Residuals

Do the residuals have a normal distribution?



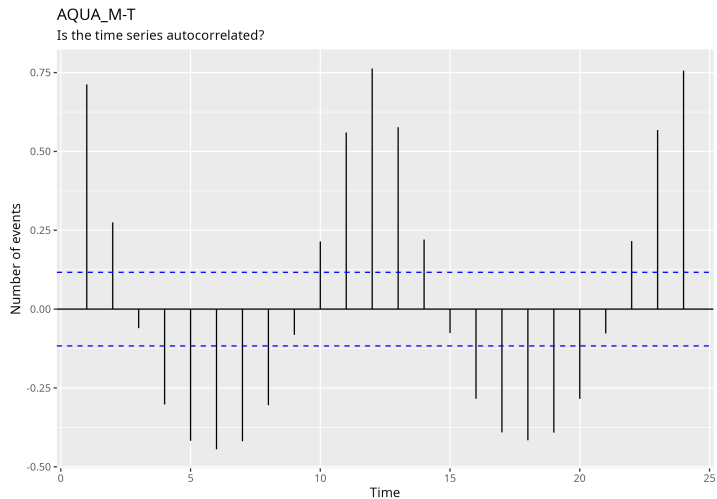
AQUA M T

Time series trend (with smoothing)



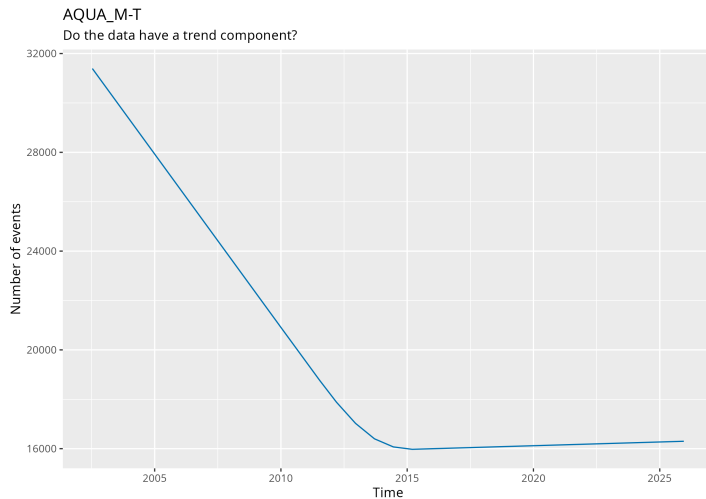
Do the data have trend when smoothed?

Time series autocorrelation



Is the time series autocorrelated?

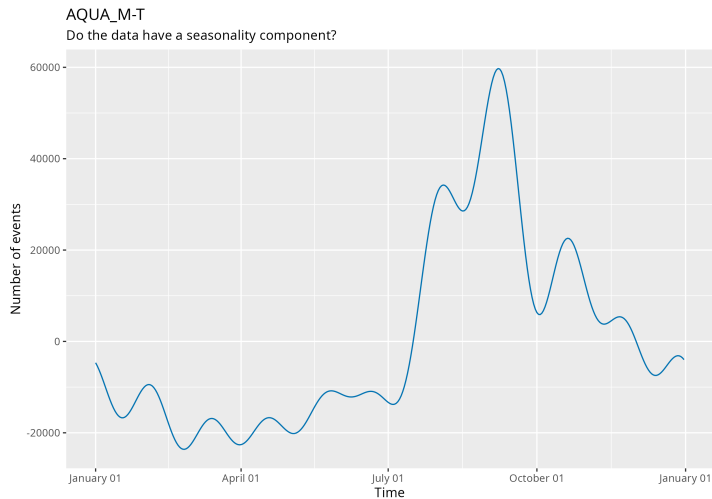
Time series trend



Do the data have a trend component?



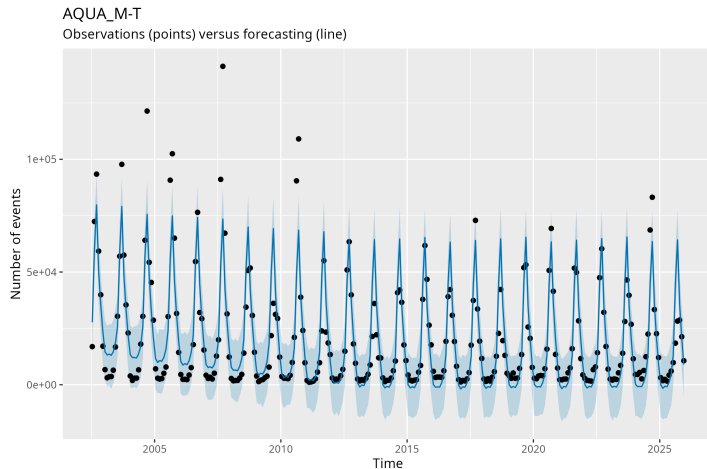
Time series seasonality



Do the data have a seasonality component?



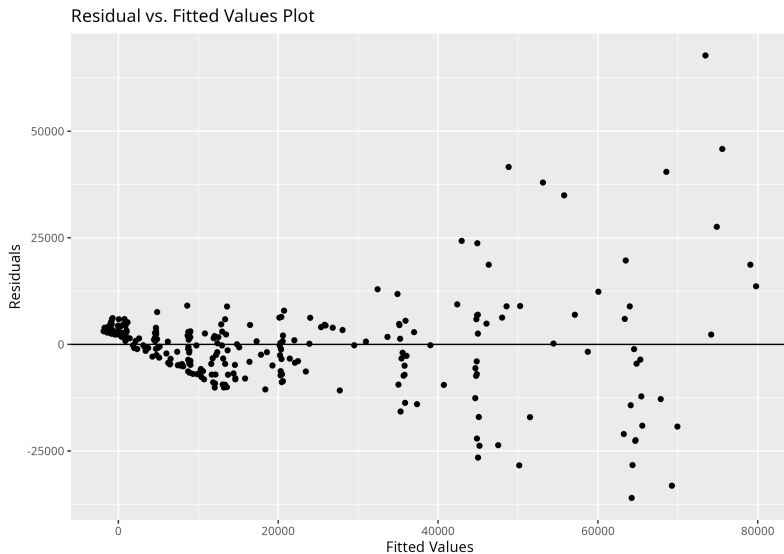
Time series versus forecast



Observations (black dots), model (blue line), and 95% confidence interval (gray shadow).



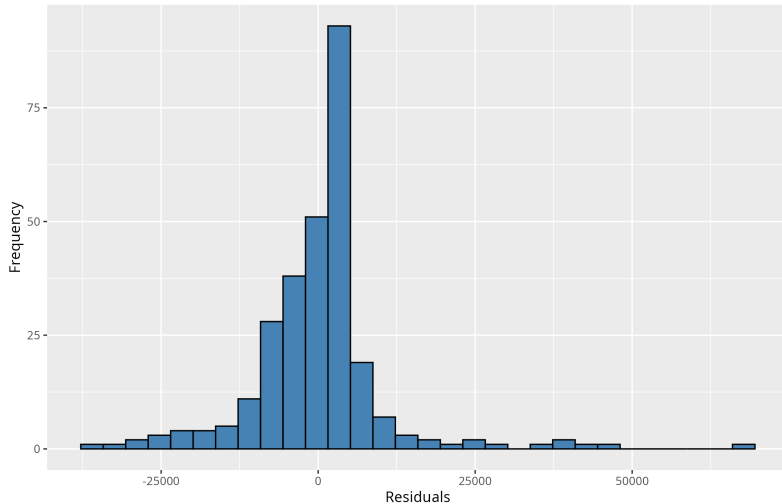
Time series versus forecast



Time series versus forecast

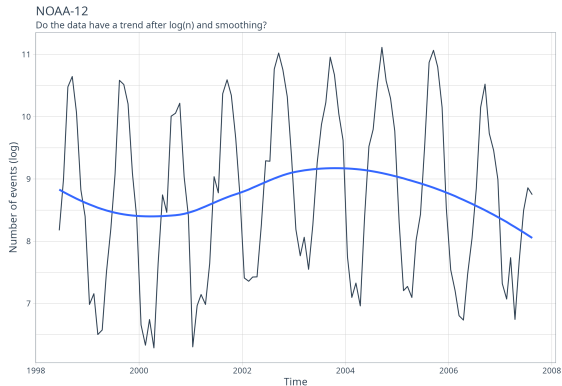
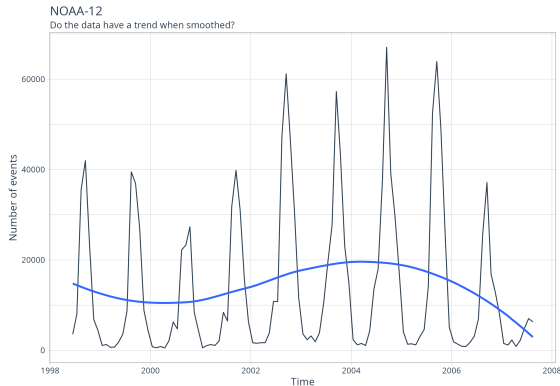
Histogram of Residuals

Do the residuals have a normal distribution?



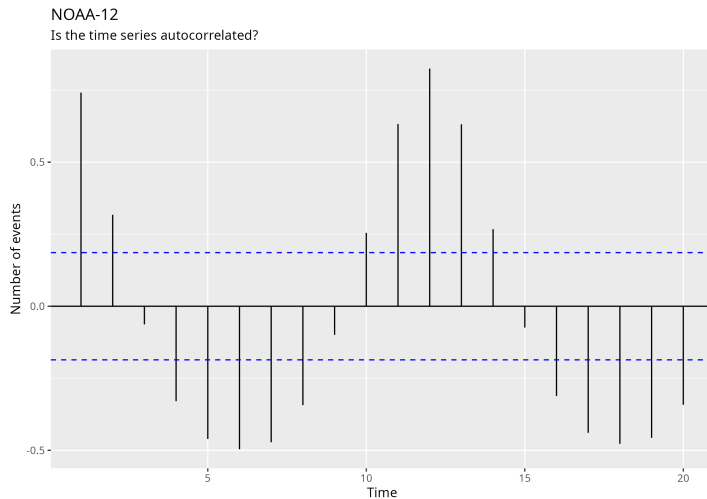
NOAA 12

Time series trend (with smoothing)



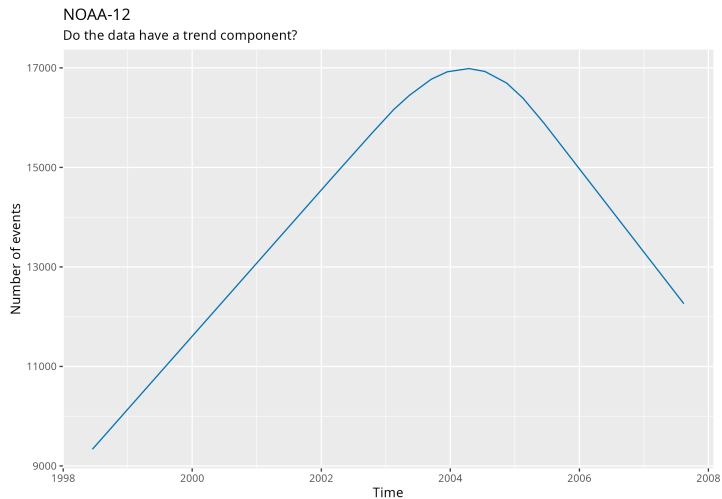
Do the data have a trend when smoothed?

Time series autocorrelation



Is the time series autocorrelated?

Time series trend



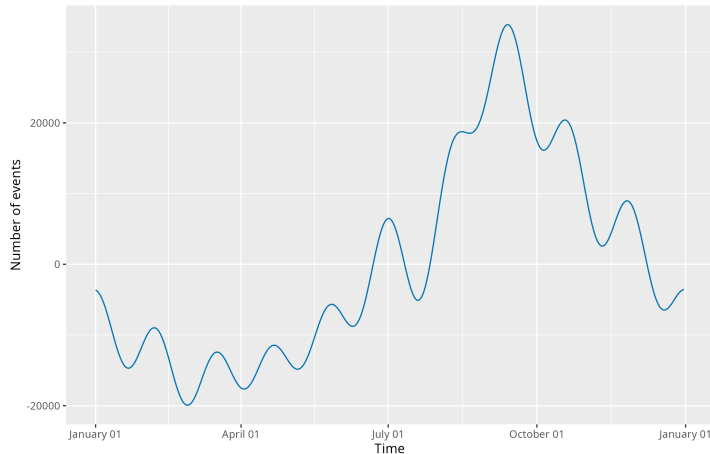
Do the data have a trend component?



Time series seasonality

NOAA-12

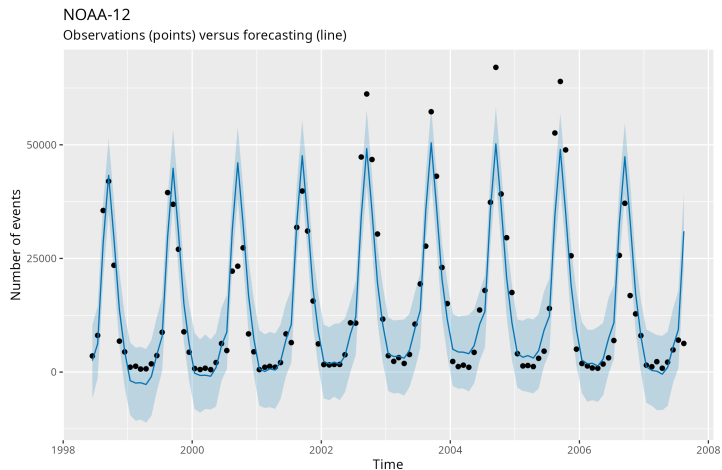
Do the data have a seasonality component?



Do the data have a seasonality component?



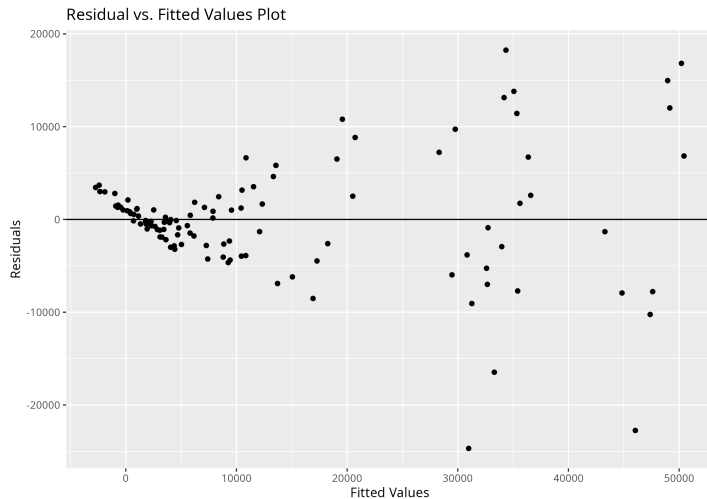
Time series versus forecast



Observations (black dots), model (blue line), and 95% confidence interval (gray shadow).

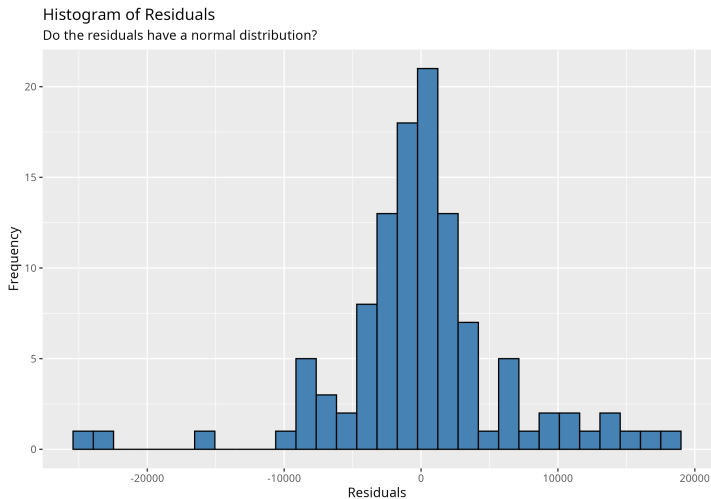


Time series versus forecast



Large fitted fire foci values have large residuals.

Time series versus forecast



Do the residuals have a normal distribution?

Summary

Name	Trend	TLog	Autocor	Season	Random Res	Res Dist
NPP-375D	+	+	12 m	yes	TODO	Normal
NPP-375-PM	+	+	12 m	yes	TODO	Normal
AQUA M T	-+	-+	12 m	yes	TODO	Normal
NOAA 12	-	+-	12 m	yes	TODO	Normal

Table: Summary of findings in in the fire foci time series.

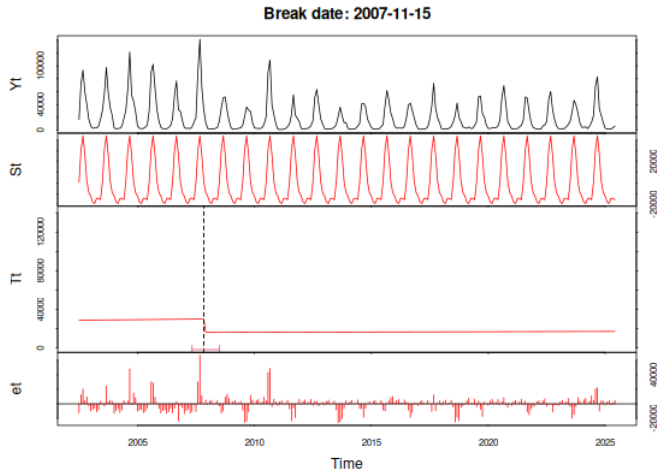
BFast

Breaks For Additive Seasonal and Trend - BFAST

- Additive decomposition model that iteratively fits a linear trend and a seasonal model [6, 7].
- Y_t is the observed data at time t .
- T_t is the trend component.
- S_t is the season component.
- e_t is the remaining variation in the data beyond the seasonal and trend components.

$$Y_t = T_t + S_t + e_t$$

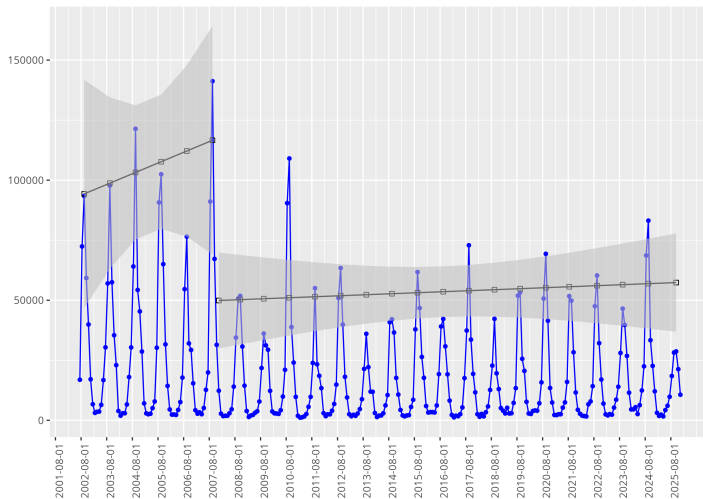
Time series breaks (BFAST) AQUA M-T



BFAST identified a break in the trend component of the AQUA M-T fire foci times series.



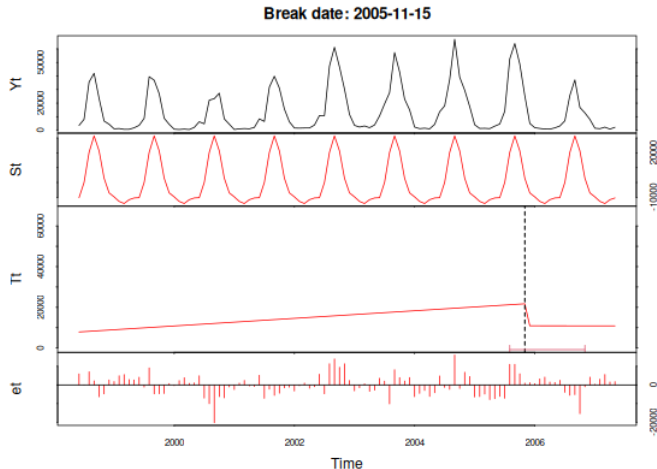
AQUA M-T annual top values trend after BFast breaks



Annual-peak trends are positive before and after the break. (in contrast to no break ).

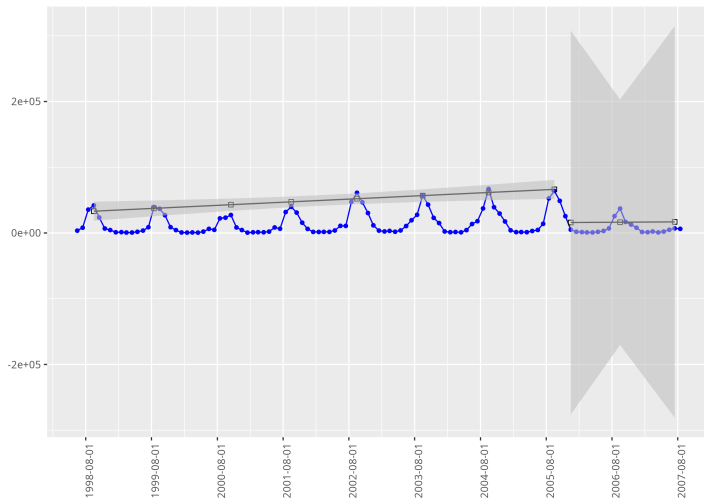


Time series breaks (BFast) NOAA-12



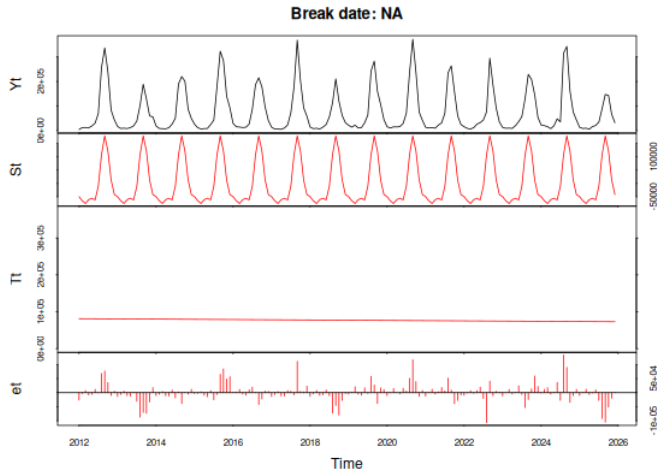
BFAST didn't find a break in NOAA-12 fire foci time series.

NOAA-12 annual top values trend after BFast breaks



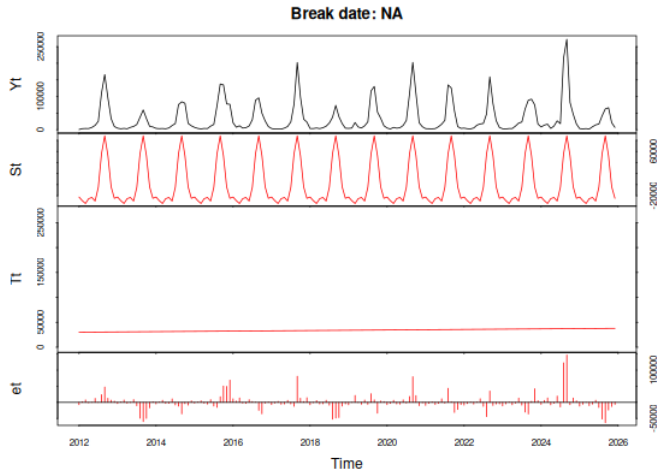
The annual-peak trend is slightly positive before break and neutral after it. (in contrast to no break).

Time series breaks (BFAST) NPP-375-PM



BFAST didn't find a break in NPP-375-PM fire foci time series.

Time series breaks (BFast) NPP-375-PM



BFAST didn't find a break in NPP-375D fire foci time series.



Queimadas method (12-month regression)

Queimadas method for forecasting heat foci

- 1 Get time series of two reference satellites:
 - ▶ One which we would like to extend in time (A.K.A. y).
 - ▶ The other one covering the whole time line we would like to forecast (A.K.A. x).
- 2 Adjust a linear model ($y = Ax + b$) to the overlapping observations in the time series of two reference satellites, using only the month of the observations.
- 3 Use the model to predict the whole time line.
- 4 Check forecast results.

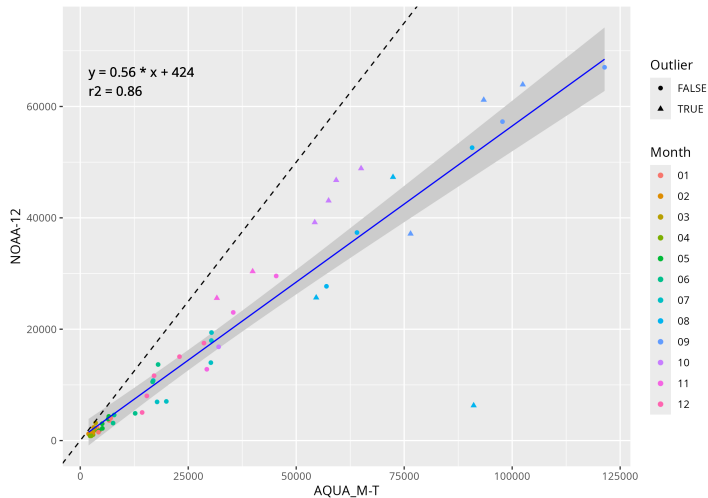


Outliers and confidence intervals

- Outliers were determined on the regression residuals.
- The outliers are outside the range given by $\pm 1.58 IQR / \sqrt{n}$, where IQR is the Interquartile range, that is, the range between the first and third quartiles.
- The confidence intervals (gray shadows in the regression and forecast figures) is given the 95% confidence level.



Linear model NOAA-12 vs. AQUA_M-T



Linear model with 95% C.I. The identity line is dashed.



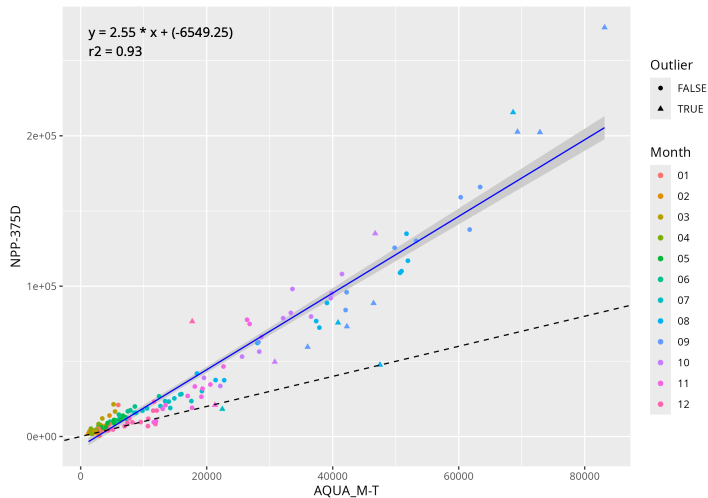
Outliers linear model NOAA-12 vs. AQUA_M-T

Year	Month	x	y	yhat	Residual
2002	08	72412	47317	41024.58	37898.51
2002	09	93417	61181	52801.84	48617.11
2002	10	59257	46764	33648.72	31109.07
2002	11	39913	30375	22802.76	20879.96
2003	10	57495	43089	32660.79	30192.19
2004	10	54292	39184	30864.90	28519.47
2005	09	102455	63933	57869.34	53204.21
2005	10	65023	48878	36881.65	34095.70
2005	11	31631	25587	18159.14	16343.94
2006	08	54630	25670	31054.42	28696.38
2006	09	76475	37137	43302.65	39980.63
2007	08	91085	6299	51494.31	47431.69

Table: Outliers of the model presented before.



Linear model NPP-375D vs. AQUA_M-T



Linear model with 95% C.I. The identity line is dashed.

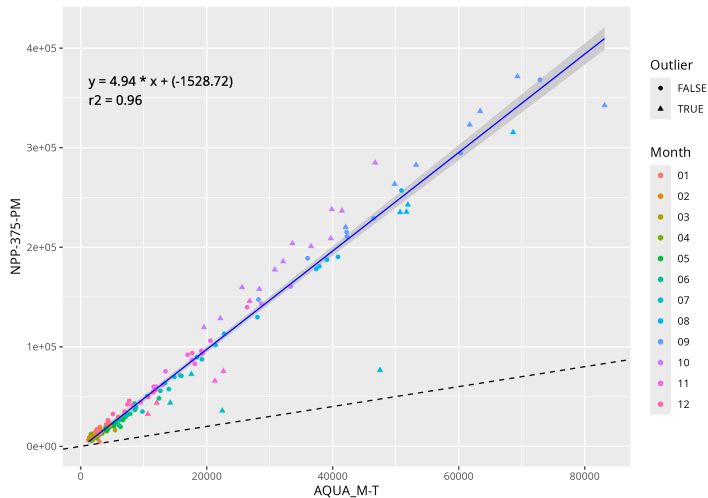


Outliers linear model NPP-375D vs. AQUA_M-T

Year	Month	x	y	yhat	Residual
2013	09	36019	59535	85270.83	82298.38
2014	08	40845	75686	97573.33	94182.70
2015	10	46741	135011	112603.49	108663.82
2015	12	17703	76526	38579.46	36598.43
2016	10	30809	49627	71989.44	69419.85
2017	09	72895	202308	179275.59	172670.03
2018	09	42251	73180	101157.53	97639.17
2020	09	69329	202617	170185.10	163954.96
2022	08	47507	47640	114556.19	110542.91
2023	09	46486	88659	111953.44	108038.17
2024	07	22478	18231	50751.95	48651.14
2024	08	68635	215589	168415.95	162258.60
2024	09	83154	272041	205427.96	197731.67
2025	11	21301	20907	47751.53	45692.14

Table: Outliers of the model presented before.

Linear model NPP-375-PM vs. AQUA_M-T



Linear model with 95% C.I. The identity line is dashed.

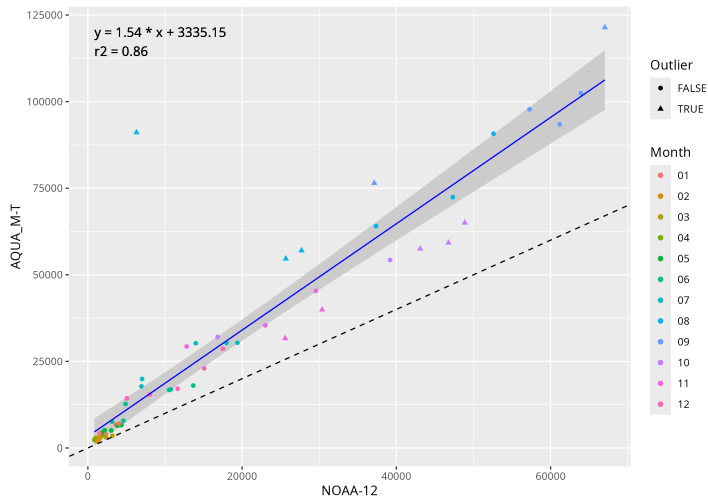


Outliers linear model NPP-375-PM vs. AQUA_M-T I

Year	Month	x	y	yhat	Residual
2012	09	63408	336595	311987.67	303878.61
2012	10	39860	238154	195556.24	190784.66
2013	10	22159	128579	108034.89	105016.89
2014	09	42049	220178	206379.60	201322.96
2014	10	36568	201003	179279.18	174918.49
2015	09	61739	323070	303735.42	295875.58
2015	10	46741	284887	229578.86	223886.79
2016	10	30809	177465	150804.21	147091.65
2017	07	17568	72520	85335.02	82474.89
2017	10	33607	203876	164638.72	160624.54
2018	10	19568	119528	95223.87	92316.61
2019	08	51935	242744	255260.23	248835.87
2019	09	53234	282749	261683.04	255071.92
2019	10	25613	159766	125112.95	121870.04
2020	08	50694	235086	249124.19	242877.03
2020	09	69329	371618	341263.64	332262.29
2020	10	41468	236591	203506.88	198526.71
2021	08	51711	235453	254152.68	247760.40
2021	09	49829	263437	244847.26	238722.83
2021	10	28342	157917	138606.30	135133.53
2022	07	14212	43692	68741.51	65880.85
2022	08	47507	76511	233366.29	227567.89
2022	10	32130	185691	157335.80	153484.36
2023	10	39692	208865	194725.58	189975.52
2023	11	26819	145959	131075.93	127736.52
2024	07	22478	35755	109612.17	106576.90
2024	08	68635	315415	337832.20	328936.02
2024	09	83154	342525	409620.38	398500.72
2024	11	22663	75577	110526.88	107481.26



Linear model AQUA_M-T vs. NOAA-12



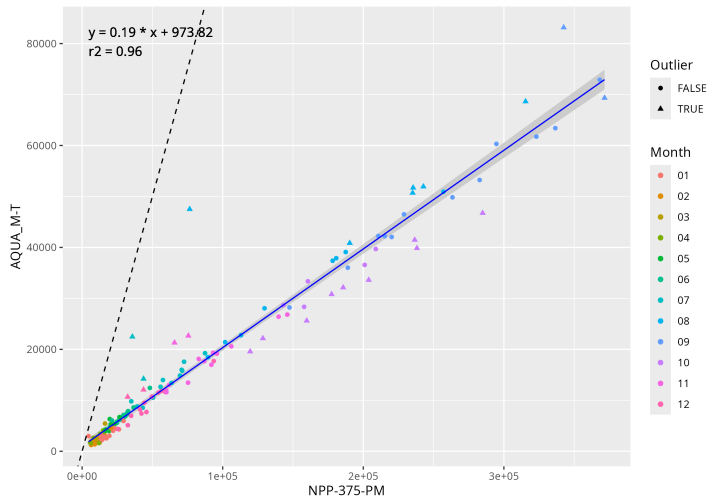
Linear model with 95% C.I. The identity line is dashed.

Outliers linear model AQUA_M-T vs. NOAA-12

Year	Month	x	y	yhat	Residual
2002	10	46764	59257	75121.65	69440.128
2002	11	30375	39913	49963.22	46240.947
2003	08	27705	57004	45864.56	42379.167
2003	10	43089	57495	69480.23	64287.930
2004	09	67038	121395	106243.87	97642.947
2005	10	48878	65023	78366.81	72395.960
2005	11	25587	31631	42613.25	39288.884
2006	08	25670	54630	42740.67	39410.484
2006	09	37137	76475	60343.43	55893.546
2007	08	6299	91085	13004.62	9603.641

Table: Outliers of the model presented before.

Linear model AQUA_M-T vs. NPP-375-PM



Linear model with 95% C.I. The identity line is dashed.



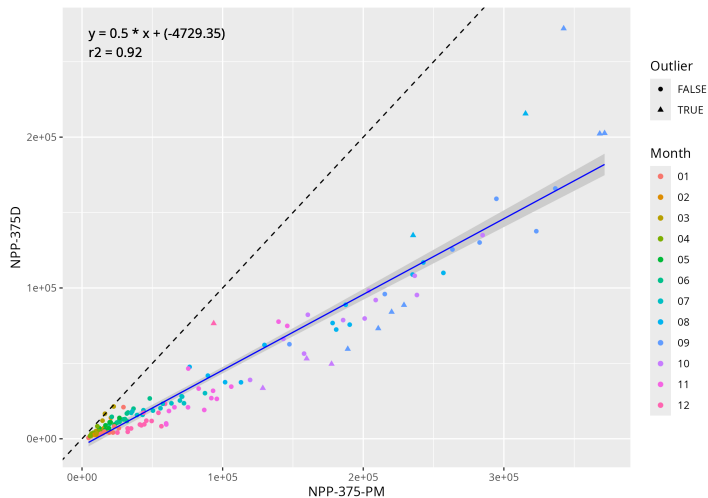
Outliers linear model AQUA_M-T vs. NPP-375-PM

Year	Month	x	y	yhat	Residual
2012	10	238154	39860	47087.642	45932.821
2013	10	128579	22159	25870.605	25221.508
2014	08	190370	40845	37835.213	36929.628
2015	10	284887	46741	56136.565	54719.203
2016	10	177465	30809	35336.415	34492.563
2017	10	203876	33607	40450.384	39477.197
2018	10	119528	19568	24118.058	23495.127
2019	08	242744	51935	47976.405	46796.422
2019	10	159766	25613	31909.352	31144.376
2020	08	235086	50694	46493.584	45355.487
2020	09	371618	69329	72930.312	71002.679
2020	10	236591	41468	46784.998	45638.707
2021	08	235453	51711	46564.647	45424.553
2022	07	43692	14212	9433.912	8832.751
2022	08	76511	47507	15788.664	15225.096
2022	10	185691	32130	36929.217	36046.370
2024	07	35755	22478	7897.069	7276.859
2024	08	315415	68635	62047.710	60453.197
2024	09	342525	83154	67297.027	65542.741
2024	11	75577	22663	15607.814	15044.175
2024	12	43682	12075	9431.976	8830.792
2025	11	65739	21301	13702.879	13134.834
2025	12	32408	10669	7248.988	6619.743

Table: Outliers of the model presented before.



Linear model NPP-375D vs. NPP-375-PM



Linear model with 95% C.I. The identity line is dashed.



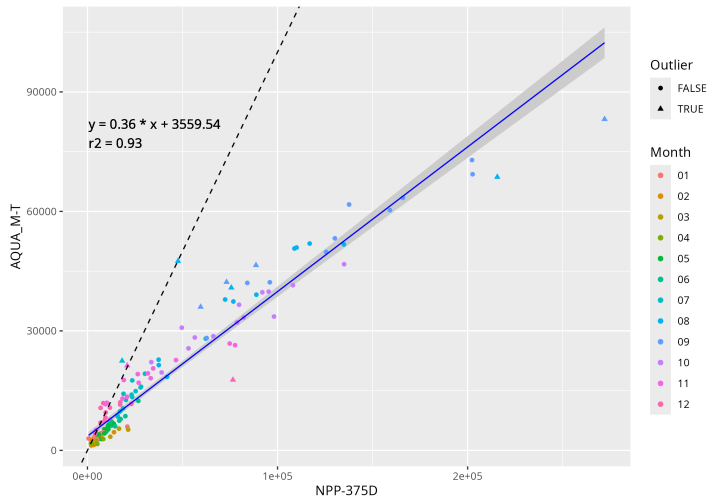
Outliers linear model NPP-375D vs. NPP-375-PM

Year	Month	x	y	yhat	Residual
2013	09	188981	59535	90176.05	86860.92
2013	10	128579	33591	59842.44	57448.30
2014	09	220178	84087	105843.04	101940.58
2015	12	93633	76526	42292.72	40179.44
2016	10	177465	49627	84392.77	81280.29
2017	09	368188	202308	180172.99	173138.78
2018	09	210659	73180	101062.64	97344.35
2019	10	159766	53159	75504.41	72682.86
2020	09	371618	202617	181895.52	174785.62
2021	08	235453	134894	113514.08	109308.94
2023	09	228974	88659	110260.35	106184.58
2024	08	315415	215589	153670.63	147789.41
2024	09	342525	272041	167285.15	160814.62

Table: Outliers of the model presented before.



Linear model AQUA_M-T vs. NPP-375D



Linear model with 95% C.I. The identity line is dashed.

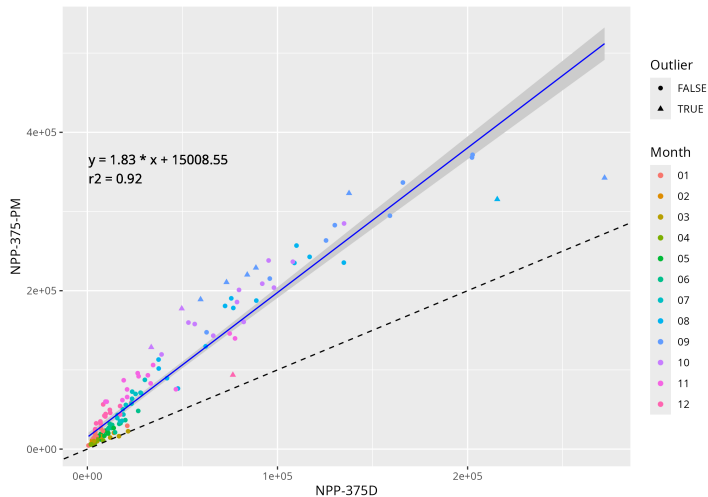


Outliers linear model AQUA_M-T vs. NPP-375D

Year	Month	x	y	yhat	Residual
2013	09	59535	36019	25179.73	24333.935
2014	08	75686	40845	31044.98	30052.502
2015	12	76526	17703	31350.03	30348.737
2018	09	73180	42251	30134.92	29168.135
2022	08	47640	47507	20860.05	20085.684
2023	09	88659	46486	35756.14	34617.960
2024	07	18231	22478	10180.14	9395.772
2024	08	215589	68635	81850.88	78893.926
2024	09	272041	83154	102351.48	98526.885
2025	11	20907	21301	11151.93	10379.892

Table: Outliers of the model presented before.

Linear model NPP-375-PM vs. NPP-375D



Linear model with 95% C.I. The identity line is dashed.



Outliers linear model NPP-375-PM vs. NPP-375D

Year	Month	x	y	yhat	Residual
2013	09	59535	188981	123768.55	119261.28
2013	10	33591	128579	76373.41	72408.77
2014	09	84087	220178	168620.75	162840.52
2015	09	137635	323070	266443.56	256879.43
2015	12	76526	93633	154808.13	149472.22
2016	10	49627	177465	105668.37	101492.36
2018	09	73180	210659	148695.58	143543.52
2023	09	88659	228974	176973.00	170907.60
2024	08	215589	315415	408851.85	393094.09
2024	09	272041	342525	511979.76	491598.32

Table: Outliers of the model presented before.

Linear regression parameters

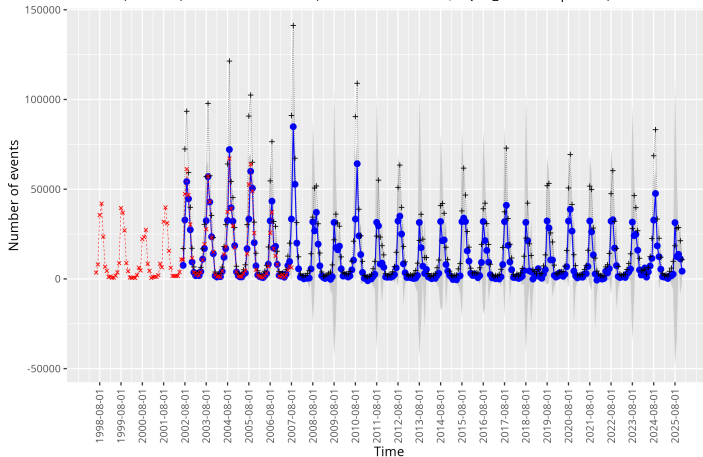
satellite_x	satellite_y	equation	r2
AQUA_M-T	NOAA-12	$y = 0.561 * x + 424.005$	0.86
AQUA_M-T	NPP-375-PM	$y = 4.944 * x + (-1528.724)$	0.96
AQUA_M-T	NPP-375D	$y = 2.549 * x + (-6549.248)$	0.93
NOAA-12	AQUA_M-T	$y = 1.535 * x + 3335.153$	0.86
NPP-375-PM	AQUA_M-T	$y = 0.194 * x + 973.819$	0.96
NPP-375-PM	NPP-375D	$y = 0.502 * x + (-4729.351)$	0.92
NPP-375D	AQUA_M-T	$y = 0.363 * x + 3559.538$	0.93
NPP-375D	NPP-375-PM	$y = 1.827 * x + 15008.547$	0.92

Table: Parameters of the linear regressions presented before.

Stretch NOAA-12 to match AQUA_M-T

Transformation of AQUA_M-T into NOAA-12

Forecast (blue line) versus observations (NOAA-12 red crosses, AQUA_M-T black pluses)



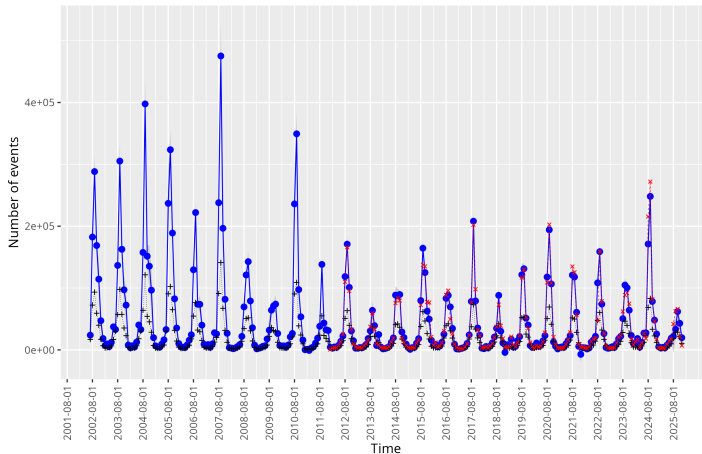
Forecast made using the regression model presented before.



Stretch NPP-375D to match AQUA_M-T

Transformation of AQUA_M-T into NPP-375D

Forecast (blue line) versus observations (NPP-375D red crosses, AQUA_M-T black plusses)

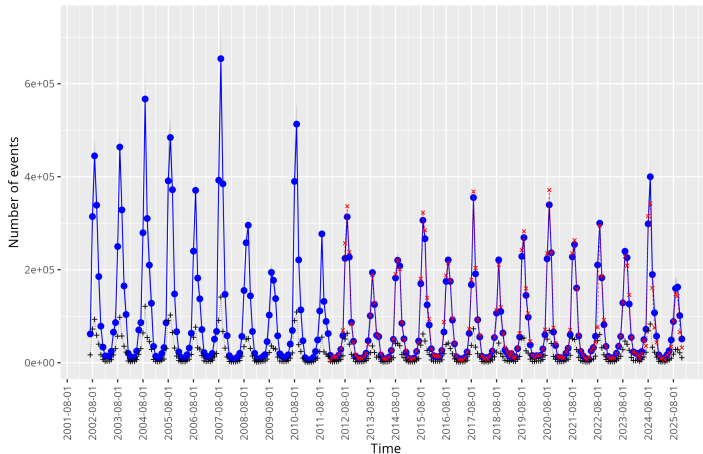


Forecast made using the regression model presented before.

Stretch NPP-375-PM to match AQUA_M-T

Transformation of AQUA_M-T into NPP-375-PM

Forecast (blue line) versus observations (NPP-375-PM red crosses, AQUA_M-T black plusses)



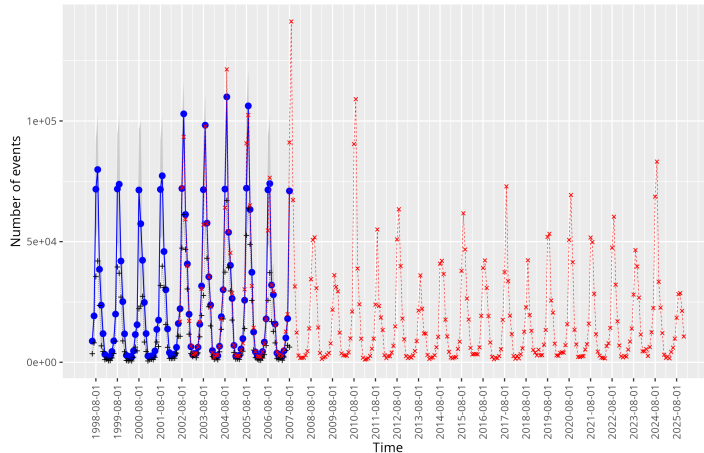
Forecast made using the regression model presented before.



Stretch match AQUA_M-T to match NOAA-12

Transformation of NOAA-12 into AQUA_M-T

Forecast (blue line) versus observations (AQUA_M-T red crosses, NOAA-12 black plusses)



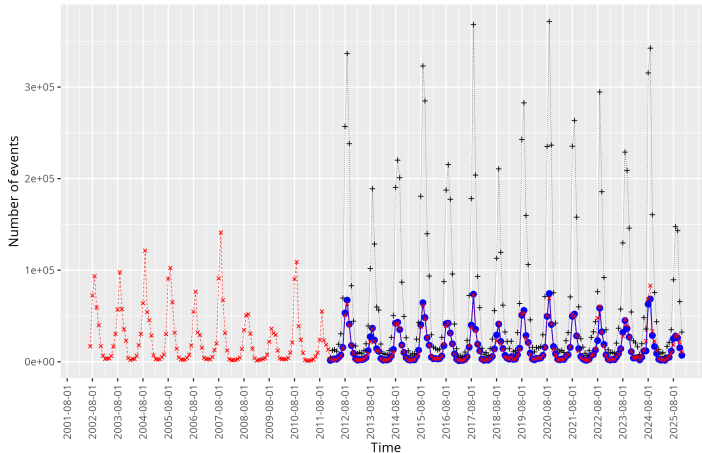
Forecast made using the regression model presented before.



Stretch match AQUA_M-T to match NPP-375-PM

Transformation of NPP-375-PM into AQUA_M-T

Forecast (blue line) versus observations (AQUA_M-T red crosses, NPP-375-PM black plusses)



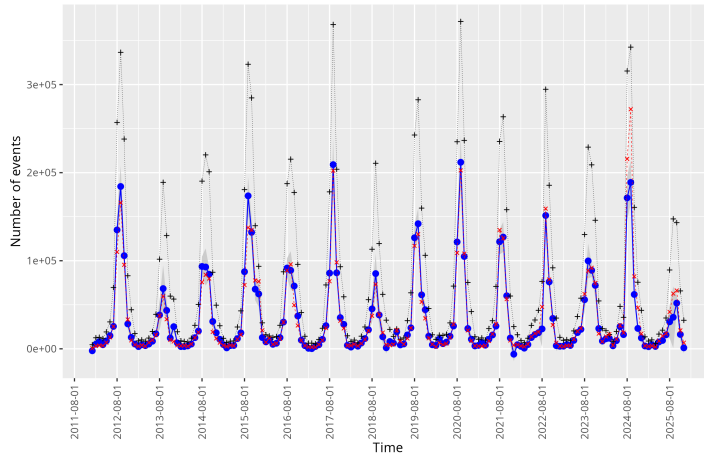
Forecast made using the regression model presented before.



Stretch match NPP-375D to match NPP-375-PM

Transformation of NPP-375-PM into NPP-375D

Forecast (blue line) versus observations (NPP-375D red crosses, NPP-375-PM black plusses)



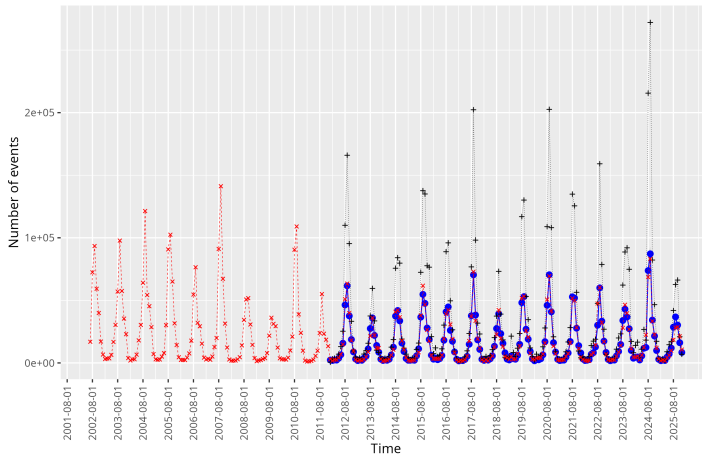
Forecast made using the regression model presented before.



Stretch match AQUA_M-T to match NPP-375D

Transformation of NPP-375D into AQUA_M-T

Forecast (blue line) versus observations (AQUA_M-T red crosses, NPP-375D black plusses)



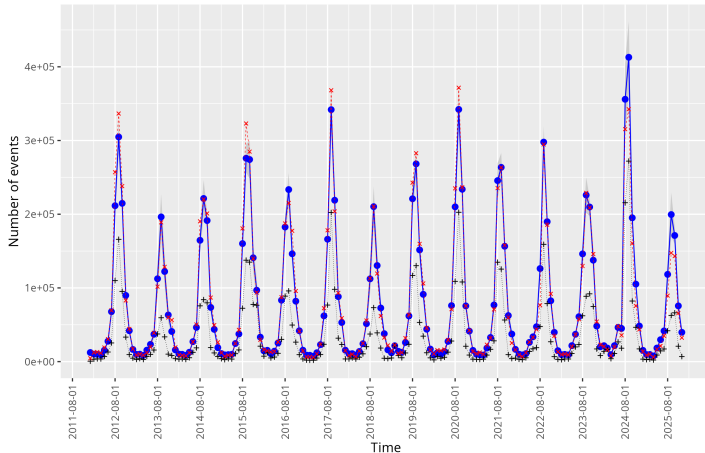
Forecast made using the regression model presented before.



Stretch match NPP-375-PM to match NPP-375D

Transformation of NPP-375D into NPP-375-PM

Forecast (blue line) versus observations (NPP-375-PM red crosses, NPP-375D black plusses)



Forecast made using the regression model presented before.

Queimadas method (1-month regression)



Queimadas (1-month) method for forecasting heat foci

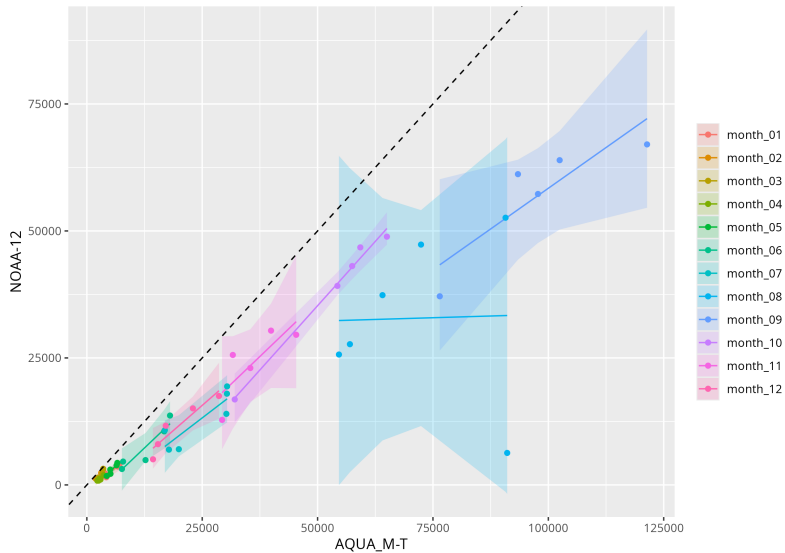
- 1 Get time series of two reference satellites:
 - ▶ One which we would like to extend in time (A.K.A. y).
 - ▶ The other one covering the whole time line we would like to forecast (A.K.A. x).
- 2 Adjust a linear model ($y = Ax + b$), in a month-by-month basis, to the overlapping observations in the time series of two reference satellites, using only the month of the observations.
- 3 Use the models to predict the whole time line.
- 4 Check forecast results.



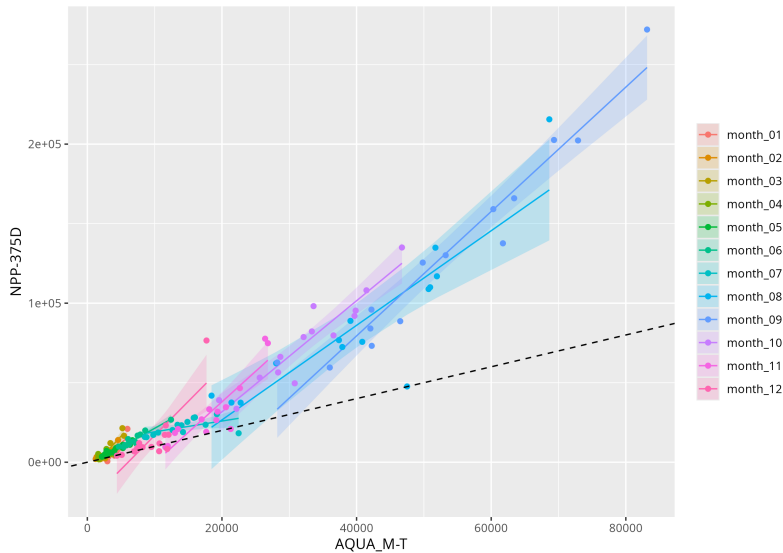
Confidence intervals

- The confidence intervals (gray shadows in the regression and forecast figures) is given at the 95% confidence level.

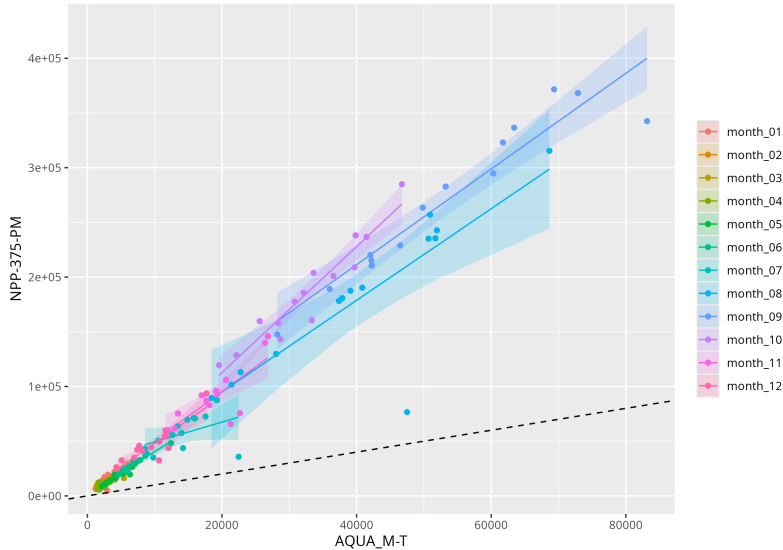
Linear model NOAA-12 vs. AQUA_M-T



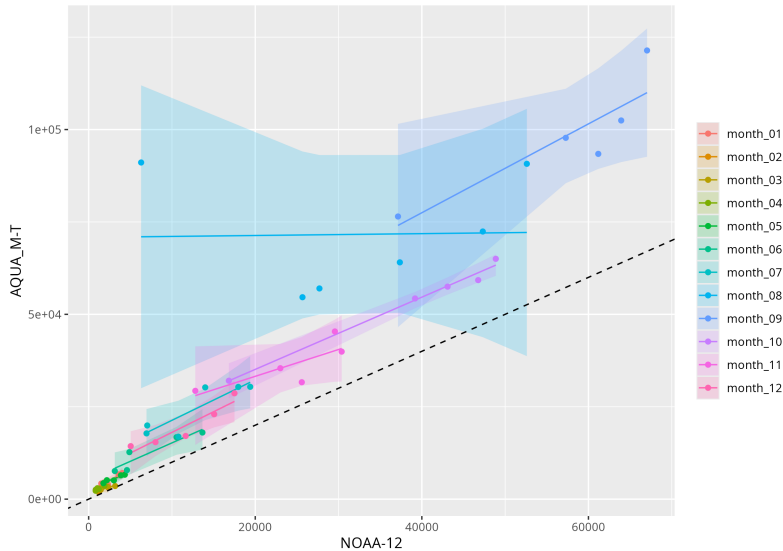
Linear model NPP-375D vs. AQUA_M-T



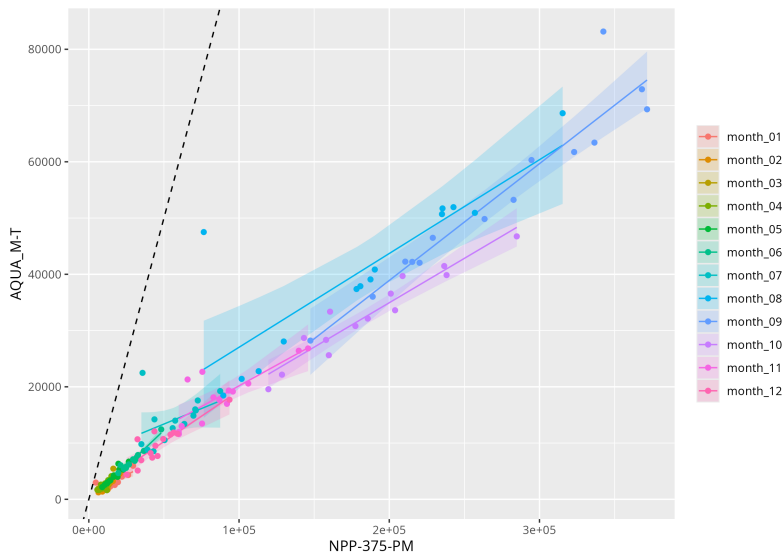
Linear model NPP-375-PM vs. AQUA_M-T



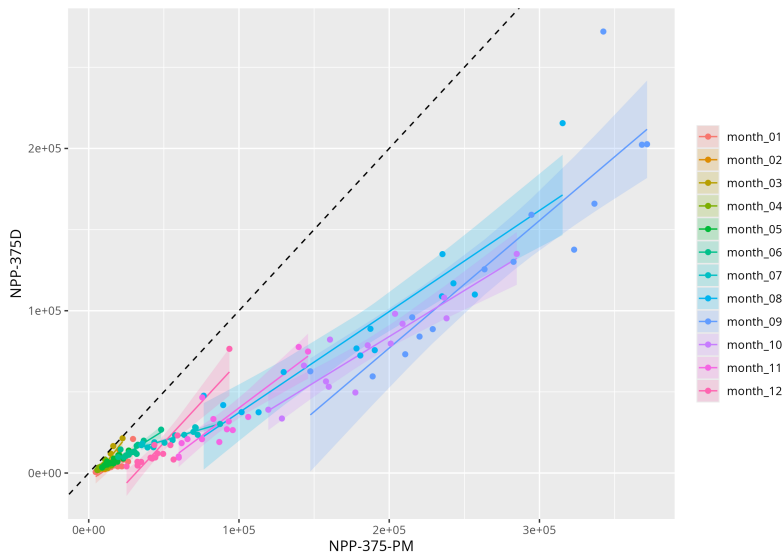
Linear model AQUA_M-T vs. NOAA-12



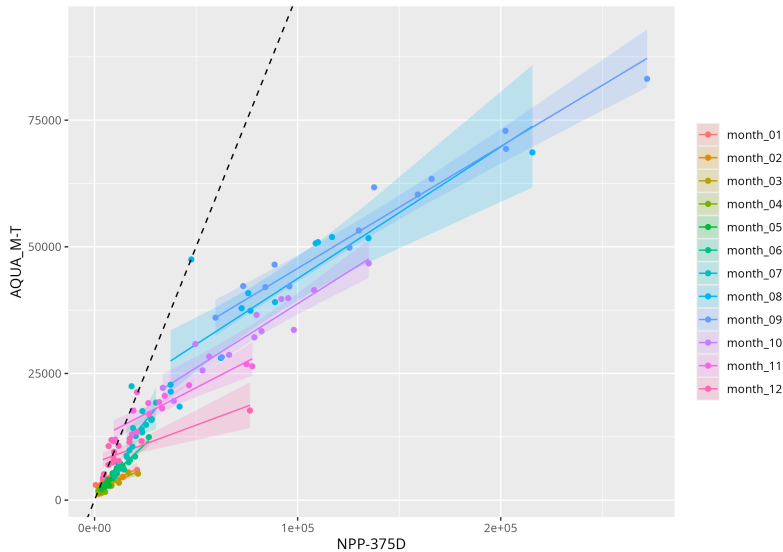
Linear model AQUA_M-T vs. NPP-375-PM



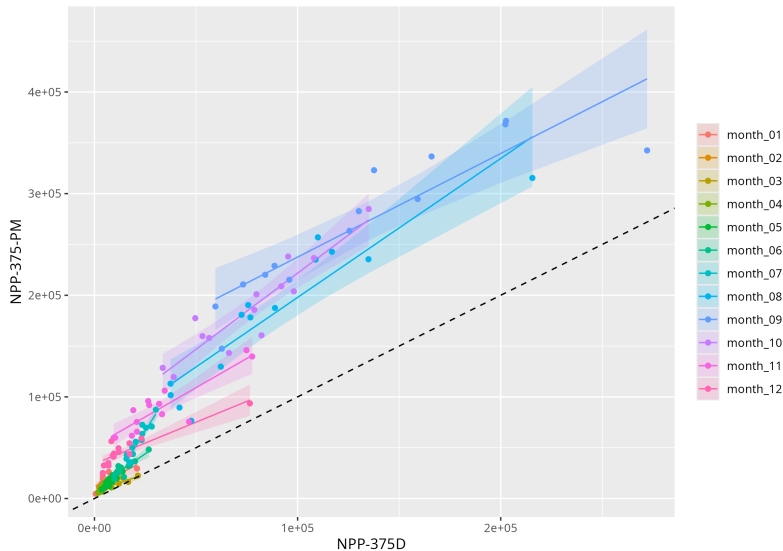
Linear model NPP-375D vs. NPP-375-PM



Linear model AQUA_M-T vs. NPP-375D



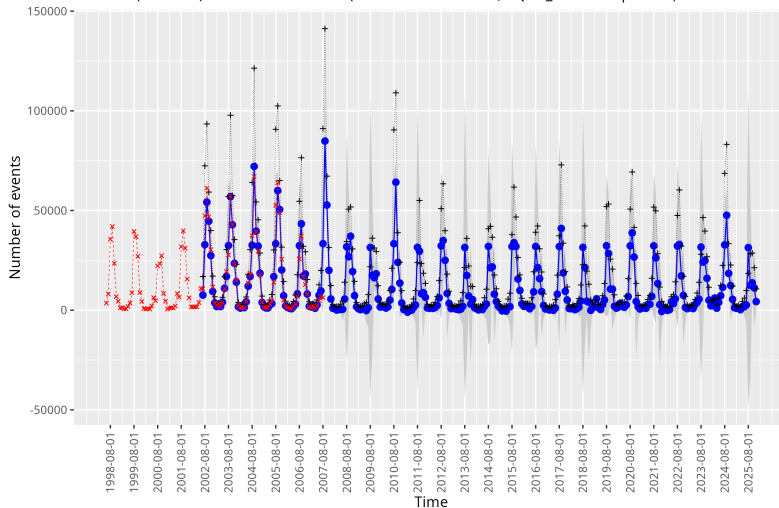
Linear model NPP-375-PM vs. NPP-375D



Stretch NOAA-12 to match AQUA_M-T (month-by-month)

Transformation of AQUA_M-T into NOAA-12

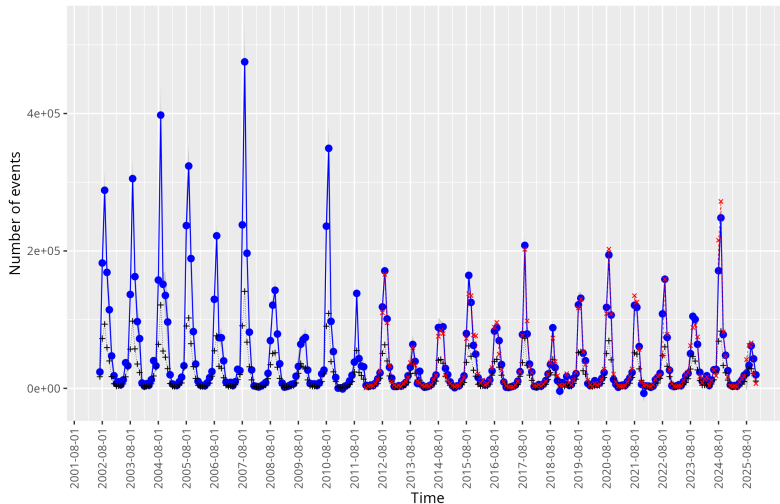
Forecast (blue line) versus observations (NOAA-12 red crosses, AQUA_M-T black plusses)



Stretch NPP-375D to match AQUA_M-T (month-by-month)

Transformation of AQUA_M-T into NPP-375D

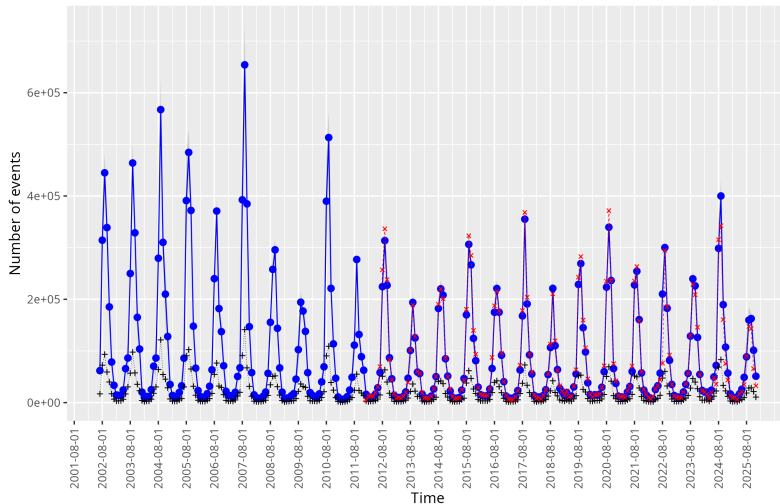
Forecast (blue line) versus observations (NPP-375D red crosses, AQUA_M-T black plusses)



Stretch NPP-375-PM to match AQUA_M-T (month-by-month)

Transformation of AQUA_M-T into NPP-375-PM

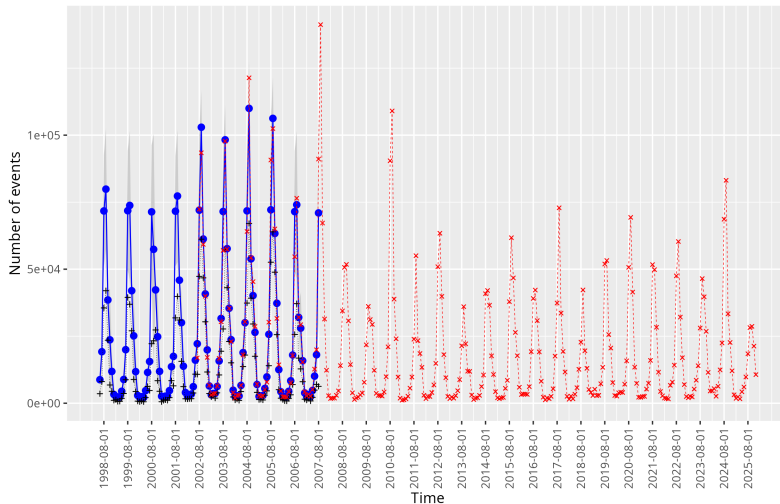
Forecast (blue line) versus observations (NPP-375-PM red crosses, AQUA_M-T black plusses)



Stretch match AQUA_M-T to match NOAA-12 (month-by-month)

Transformation of NOAA-12 into AQUA_M-T

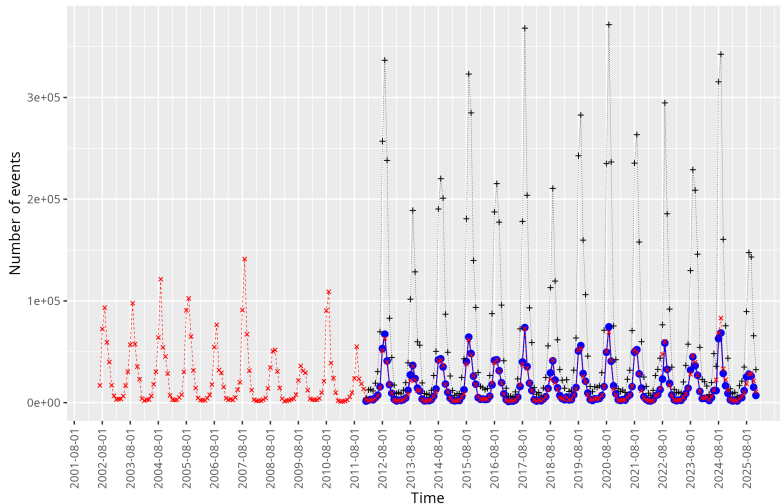
Forecast (blue line) versus observations (AQUA_M-T red crosses, NOAA-12 black plusses)



Stretch match AQUA_M-T to match NPP-375-PM (month-by-month)

Transformation of NPP-375-PM into AQUA_M-T

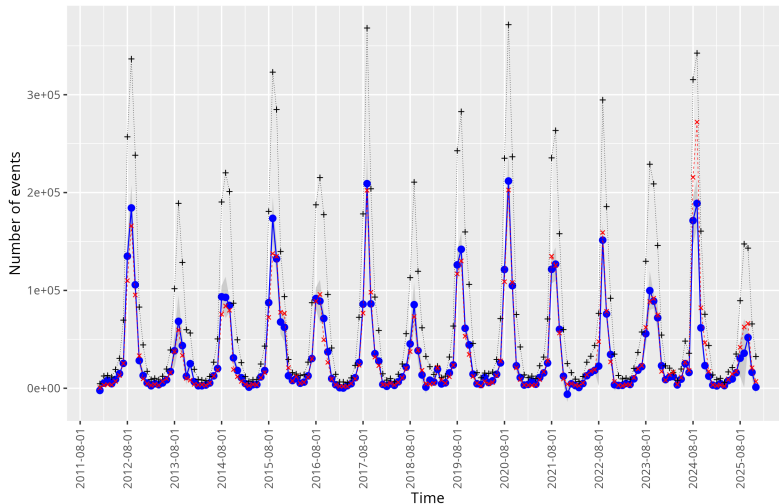
Forecast (blue line) versus observations (AQUA_M-T red crosses, NPP-375-PM black plusses)



Stretch match NPP-375D to match NPP-375-PM (month-by-month)

Transformation of NPP-375-PM into NPP-375D

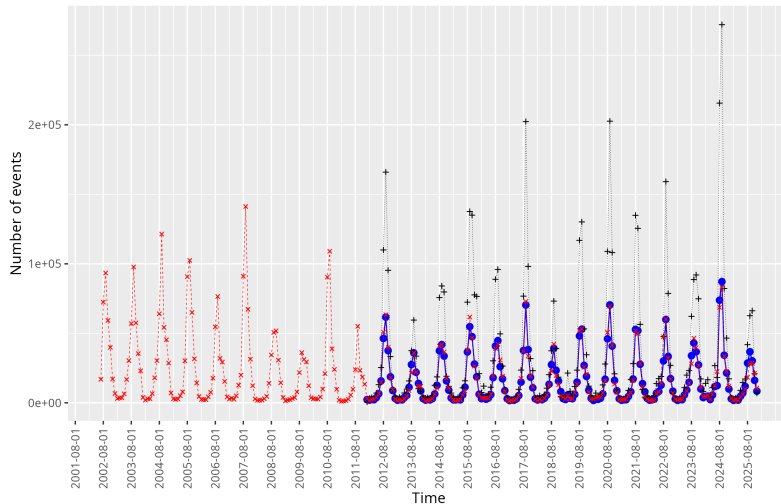
Forecast (blue line) versus observations (NPP-375D red crosses, NPP-375-PM black plusses)



Stretch match AQUA_M-T to match NPP-375D (month-by-month)

Transformation of NPP-375D into AQUA_M-T

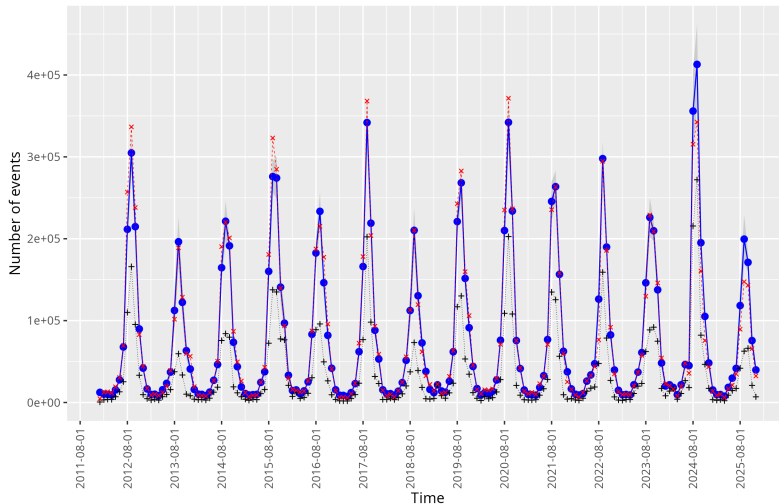
Forecast (blue line) versus observations (AQUA_M-T red crosses, NPP-375D black plusses)



Stretch match NPP-375-PM to match NPP-375D (month-by-month)

Transformation of NPP-375D into NPP-375-PM

Forecast (blue line) versus observations (NPP-375-PM red crosses, NPP-375D black plusses)

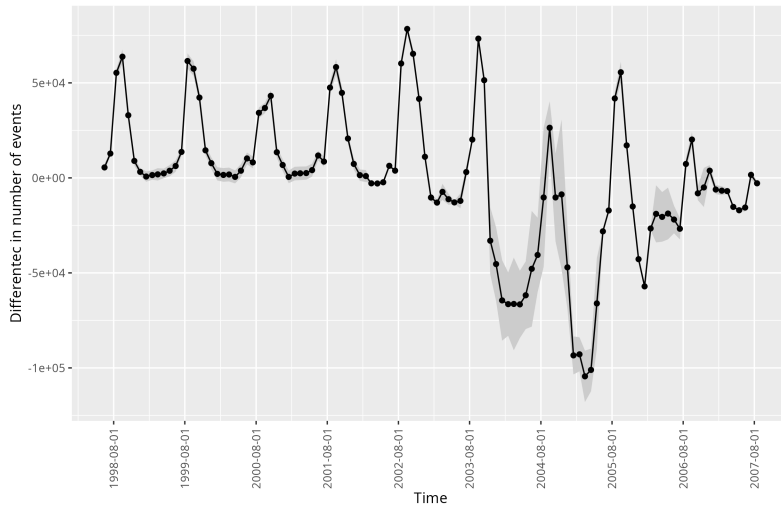


Forecast differences between Queimadas' 12 versus 1 month method

Forecast differences NOAA-12 vs. AQUA_M-T

Forecast differences using the Queimadas method 12 versus 1 month

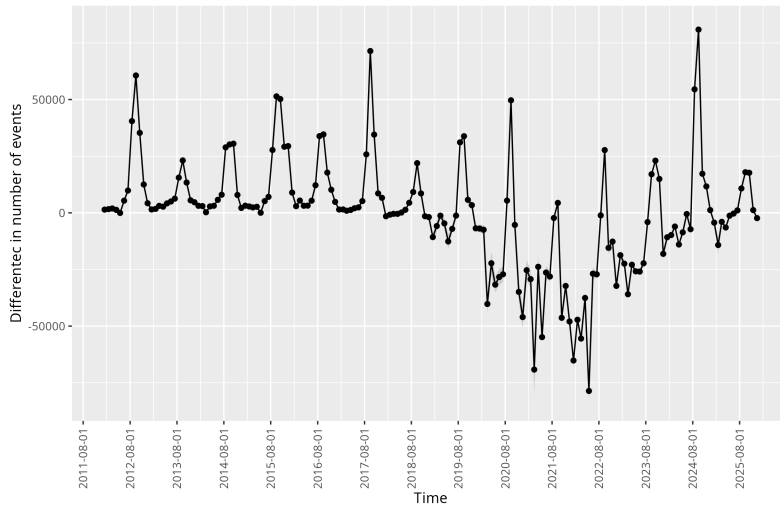
Differences in forecasts of AQUA_M-T using NOAA-12



Forecast differences NPP-375D vs. AQUA_M-T

Forecast differences using the Queimadas method 12 versus 1 month

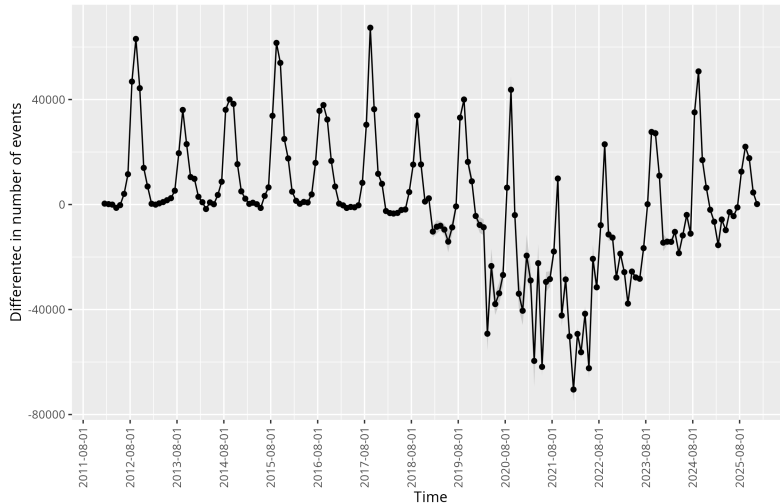
Differences in forecasts of AQUA_M-T using NPP-375D



Forecast differences NPP-375PM vs. AQUA_M-T

Forecast differences using the Queimadas method 12 versus 1 month

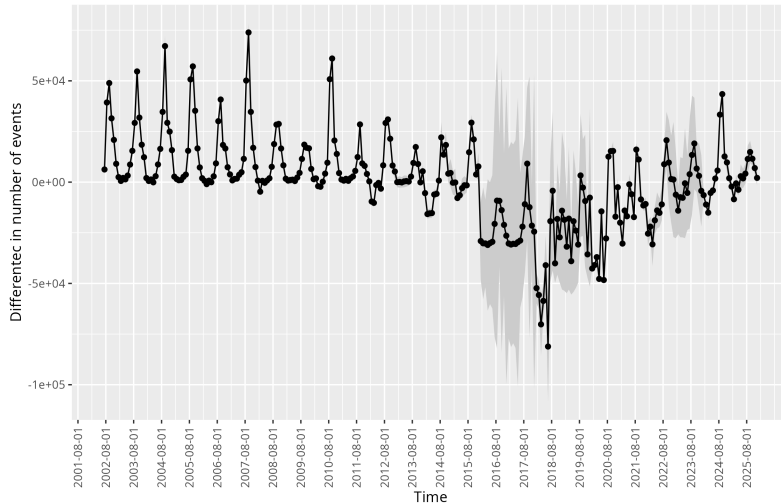
Differences in forecasts of AQUA_M-T using NPP-375-PM



Forecast differences AQUA_M-T vs. NOAA-12

Forecast differences using the Queimadas method 12 versus 1 month

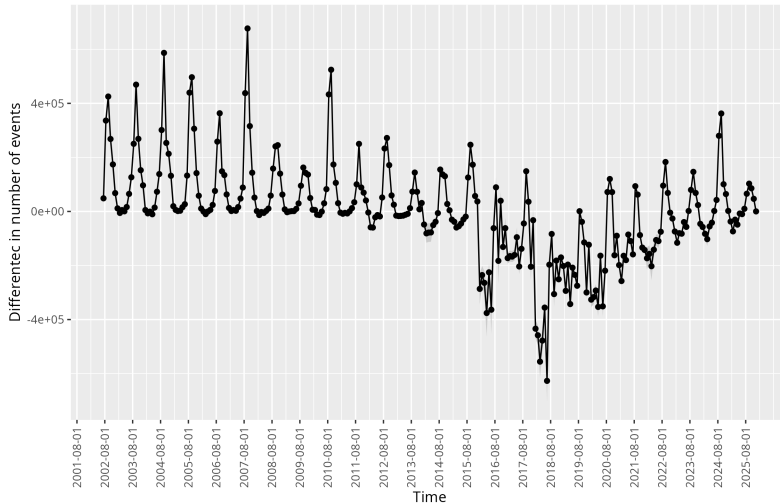
Differences in forecasts of NOAA-12 using AQUA_M-T



Forecast differences AQUA_M-T vs. NPP-375PM

Forecast differences using the Queimadas method 12 versus 1 month

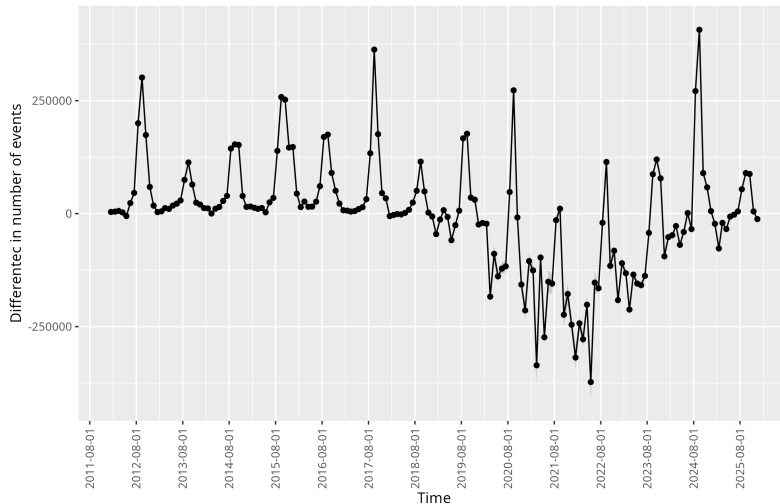
Differences in forecasts of NPP-375-PM using AQUA_M-T



Forecast differences NPP-375D vs. NPP-375PM

Forecast differences using the Queimadas method 12 versus 1 month

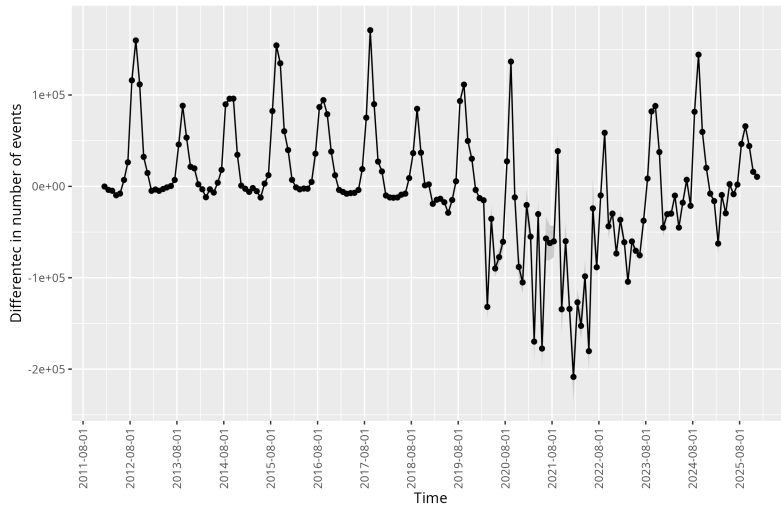
Differences in forecasts of NPP-375-PM using NPP-375D



Forecast differences NPP-375PM vs. NPP-375D

Forecast differences using the Queimadas method 12 versus 1 month

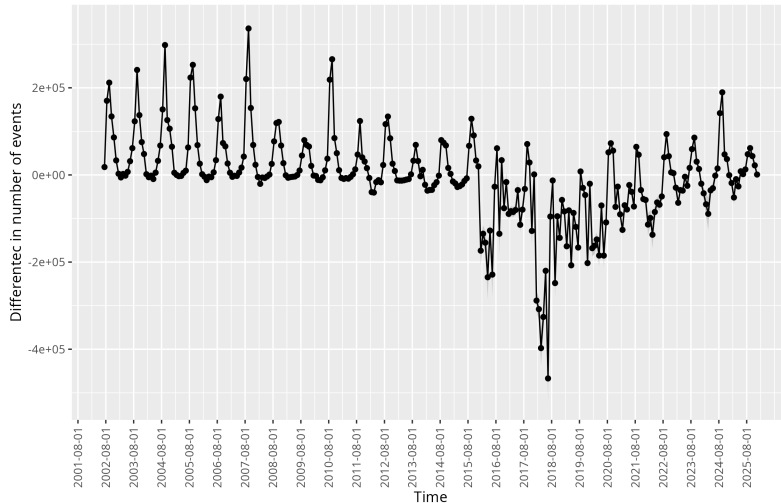
Differences in forecasts of NPP-375D using NPP-375-PM



Forecast differences AQUA_M-T vs. NPP-375D

Forecast differences using the Queimadas method 12 versus 1 month

Differences in forecasts of NPP-375D using AQUA_M-T



Future Work



Future work

- Use a curve instead of a line for regressions. NOTE: This could have been superseded by using month-by-month regressions.
- Get data from before 2003. It's not available at Queimadas' portal:
https://dataserver-coids.inpe.br/queimadas/queimadas/focos/csv/anual/Brasil_todos_sats
- Use Mann-Kendall to identify breakpoints and adjust models to each segment. The *segmented* package has an implementation **??**. NOTE: We used BFAST instead.
- Try segmented regression on time series. NOTE: We used BFAST instead.
- Spatial analysis:
 - ▶ Use Landsat grid for spatial aggregation.
 - ▶ Try bfastspatial [**bfastspatial**].



References

References I

- [1] A. C. Harvey and S. Peters. “Estimation Procedures for Structural Time Series Models”. In: *Journal of Forecasting* 9.2 (Mar. 1990), pp. 89–108. DOI: [10.1002/for.3980090203](https://doi.org/10.1002/for.3980090203).
- [2] National Institute for Space Research in Brazil. *Banco de Dados de queimadas*. 2023. URL: <http://www.inpe.br/queimadas/bdqueimadas> (visited on 02/11/2026).
- [3] A. Setzer. *Programa Queimadas, INPE, Perguntas frequentes*. Webpage. INPE, Sept. 25, 2020. URL: https://terrabrasilis.dpi.inpe.br/queimadas/portal/pages/secao_informacoes/faq/index.html (visited on 06/07/2026).
- [4] Alberto Setzer, Fabiano Morelli, and Jean Carlos Souza. “O Banco de Dados de Queimadas Do INPE”. In: *Biodiversidade Brasileira* 9.1 (2019), pp. 239–239.
- [5] Sean J. Taylor and Benjamin Letham. “Forecasting at Scale”. In: *The American Statistician* 72.1 (Jan. 2, 2018), pp. 37–45. DOI: [10.1080/00031305.2017.1380080](https://doi.org/10.1080/00031305.2017.1380080).



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- [6] Jan Verbesselt et al. “Detecting Trend and Seasonal Changes in Satellite Image Time Series”. In: *Remote Sensing of Environment* 114.1 (Jan. 2010), pp. 106–115. ISSN: 00344257. DOI: [10.1016/j.rse.2009.08.014](https://doi.org/10.1016/j.rse.2009.08.014). PMID: 16531038.
- [7] Jan Verbesselt et al. “Phenological Change Detection While Accounting for Abrupt and Gradual Trends in Satellite Image Time Series”. In: *Remote Sensing of Environment* 114.12 (Dec. 2010), pp. 2970–2980. ISSN: 00344257. DOI: [10.1016/j.rse.2010.08.003](https://doi.org/10.1016/j.rse.2010.08.003). PMID: 11818969.

